

Seminar on Algebraic Geometry 3
Exposé IV: Topologies and Sheaves
English Translation and TeX Reconstruction

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Exposé IV

Topologies and Sheaves

by M. Demazure*

This exposé is intended to acquaint the reader with the essentials of the language of topologies and sheaves (without cohomology), a language particularly convenient for questions concerning the formation of quotients, among others.⁰

The first three sections develop the language of forming quotients. The fourth, which is the central part, presents the theory of sheaves, oriented mainly toward its application to questions about quotients; the fifth applies this theory to the formation of quotients in groups and to principal homogeneous bundles. The final section concerns the category of schemes more specifically and defines several useful topologies on that category.

The reader will profitably consult [AS], [MA], [D], and SGA 4; in particular, [D] for the applications of topologies to descent theory, and SGA 4 for questions concerning universes (which are treated especially poorly in this exposé).

1. Universal Effective Epimorphisms

Throughout the remainder of this exposé, a category $\mathcal{C}$ is fixed.

Definition 1.1. A morphism $u : T \rightarrow S$ is called an epimorphism if, for every object X , the corresponding map

$$X(S) = \text{Hom}(S, X) \longrightarrow X(T) = \text{Hom}(T, X)$$

is injective.¹ It is called a universal epimorphism if, for every morphism $S' \rightarrow S$, the fiber product $T' = T \times_S S'$ exists and $u' : T' \rightarrow S'$ is an epimorphism.

Definition 1.2. A diagram of maps of sets

$$A \xrightarrow{u} B \begin{matrix} \xrightarrow{v_1} \\ \xrightarrow{v_2} \end{matrix} C$$

is said to be exact if u is injective and its image consists of those elements $b \in B$ for which $v_1(b) = v_2(b)$. A diagram of the same form in $\mathcal{C}$ is said to be exact if, for every object X of $\mathcal{C}$, the corresponding diagram of sets

$$A(X) \longrightarrow B(X) \rightrightarrows C(X)$$

*This text develops the substance of two lectures given by A. Grothendieck, supplementing them at several important points that had been passed over in silence or only briefly touched upon.

⁰Editor's note: version of October 13, 2024.

¹Editor's note: that is, if u is right-cancellable.

is exact; one then also says that u makes A a kernel of the pair of arrows (v_1, v_2) . Dually, a diagram in $\mathcal{C}$

$$C \begin{array}{c} \xrightarrow{v_1} \\ \xrightarrow{v_2} \end{array} B \xrightarrow{u} A$$

is said to be exact if it is exact as a diagram in the opposite category $\mathcal{C}^\circ$; that is, if for every object X of $\mathcal{C}$, the corresponding diagram of sets

$$X(A) \longrightarrow X(B) \rightrightarrows X(C)$$

is exact.² One also says that u makes A a cokernel of the pair of arrows (v_1, v_2) .

Definition 1.3. A morphism $u : T \rightarrow S$ is called an effective epimorphism if the fiber square $T \times_S T$ exists and the diagram

$$T \times_S T \begin{array}{c} \xrightarrow{\text{pr}_1} \\ \xrightarrow{\text{pr}_2} \end{array} T \xrightarrow{u} S$$

is exact; that is, if u makes S a cokernel of $(\text{pr}_1, \text{pr}_2)$. It is called a universal effective epimorphism if, for every morphism $S' \rightarrow S$, the fiber product $T' = T \times_S S'$ exists and the morphism $u' : T' \rightarrow S'$ is an effective epimorphism.

There are evidently implications

$$\begin{array}{ccc} \text{universal effective epimorphism} & \implies & \text{effective epimorphism} \\ \Downarrow & & \Downarrow \\ \text{universal epimorphism} & \implies & \text{epimorphism,} \end{array}$$

but in general no other implication holds.³

Definition 1.4.0.⁴ We “recall” that a morphism $u : T \rightarrow S$ is said to be *squarable* if, for every morphism $S' \rightarrow S$, the fiber product $T \times_S S'$ exists.

Lemma 1.4. Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then:

- (a) if u and v are epimorphisms, then uv is an epimorphism; and if uv is an epimorphism, then u is an epimorphism;
- (b) if u and v are universal epimorphisms, then uv is a universal epimorphism; and if uv is a universal epimorphism and u is squarable, then u is a universal epimorphism.

Lemma 1.4 follows immediately from the definitions. We deduce:

Corollary 1.5. Let $u : X \rightarrow Y$ and $u' : X' \rightarrow Y'$ be universal epimorphisms, and suppose that $Y \times Y'$ exists. Then $X \times X'$ exists and

$$u \times u' : X \times X' \longrightarrow Y \times Y'$$

is a universal epimorphism.

²Editor's note: this implies, in particular, that u is an epimorphism.

³Editor's note: for example, if $\mathcal{C} = \mathbf{Sch}$ is the category of schemes, it is easy to see that every universal epimorphism is surjective. Let

$$T = \coprod_{p \text{ prime}} \text{Spec}(\mathbf{F}_p), \quad S = \text{Spec}(\mathbf{Z}).$$

Then the morphism $u : T \rightarrow S$ is an epimorphism that is not universal. On the other hand, $T \times_S T$ identifies with T , so the two projections $T \times_S T \rightarrow T$ coincide. Since id_T does not descend to a morphism $S \rightarrow T$, this shows that u is not an effective epimorphism.

⁴Editor's note: the numbering 1.4.0 has been added for later reference.

We also record the following.

Definition 1.6.0.⁵ An object S of $\mathcal{C}$ is called *squarable* if its product with every object of $\mathcal{C}$ exists. (If $\mathcal{C}$ has a final object e , this is equivalent to saying that the morphism $S \rightarrow e$ is squarable; cf. 1.4.0.)

Lemma 1.6. *Let $u : X \rightarrow Y$ be a morphism in $\mathcal{C}/_S$. For it to be an epimorphism (respectively, a universal epimorphism, an effective epimorphism, or a universal effective epimorphism), it is sufficient that the corresponding morphism in $\mathcal{C}$ have the same property; it is also necessary if S is assumed to be a squarable object of $\mathcal{C}$.*

The proof is immediate and is left to the reader. The hypothesis that S is squarable is used to interpret the $\mathcal{C}$ -morphisms from an object Y of $\mathcal{C}/_S$ to an object Z of $\mathcal{C}$ as the $\mathcal{C}/_S$ -morphisms from Y to $Z \times S$.

Lemma 1.7. *With the notation of 1.4, if u and v are effective epimorphisms and v is a universal epimorphism, then uv is an effective epimorphism.*

To see this, consider the diagram

$$\begin{array}{ccccc}
 S & \xleftarrow{u} & T & \rightleftharpoons & T \times_S T \\
 & & \uparrow v & & \uparrow v \times_S v \\
 & & U & \rightleftharpoons & U \times_S U \\
 & & \uparrow & \nearrow & \\
 & & U \times_T U & &
 \end{array} \tag{1.7.1}$$

By hypothesis, the first row and the first column are exact; moreover, by 1.5 and 1.6, $v \times_S v$ is an epimorphism, since v is a universal epimorphism. The conclusion follows by an evident diagram chase. Indeed, if an element of $X(U)$ has the same two images in $X(U \times_S U)$, then it has, *a fortiori*, the same two images in $X(U \times_T U)$, and hence comes from an element of $X(T)$, since the first column is exact.

Since the first row is exact, it remains only to verify that the element in question has the same two images in $X(T \times_S T)$. Because $v \times_S v$ is an epimorphism, it is enough to verify that their images in $X(U \times_S U)$ are the same, which is indeed the case.

Proposition 1.8. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. If u and v are universal effective epimorphisms, then uv is a universal effective epimorphism. If uv is a universal effective epimorphism and u is squarable, then u is a universal effective epimorphism.*

The first implication follows at once from 1.7. For the second, consider the following diagram (of “bisimplicial” type):

$$\begin{array}{ccccc}
 S & \xleftarrow{u} & T & \rightleftharpoons & T \times_S T \\
 \uparrow uv & & \uparrow & & \uparrow \\
 U & \longleftarrow & U \times_S T & \rightleftharpoons & U \times_S T \times_S T \\
 \uparrow & & \uparrow & & \uparrow \\
 U \times_S U & \longleftarrow & U \times_S U \times_S T & \rightleftharpoons & U \times_S U \times_S T \times_S T.
 \end{array} \tag{1.8.1}$$

The first, second, and third columns are exact by the hypothesis that uv is a universal effective epimorphism. The second row is exact because $U \times_S T \rightarrow U$ is an effective epimorphism (it has a section over U), and the same is true of the third row, for the same reason. An evident diagram chase then shows that the first row is exact, that is, that u is an effective epimorphism. Since the hypotheses are invariant under an arbitrary base change $S' \rightarrow S$, it follows that u is in fact a universal effective epimorphism.

⁵Editor's note: the numbering 1.6.0 has been added for later reference.

Corollary 1.9. *Let $u : X \rightarrow Y$ and $u' : X' \rightarrow Y'$ be universal effective epimorphisms, and suppose that $Y \times Y'$ exists. Then $X \times X'$ exists and*

$$u \times u' : X \times X' \longrightarrow Y \times Y'$$

is a universal effective epimorphism.

The proof is the same as for 1.5, using the diagram

$$\begin{array}{ccccc}
 & & Y' & \xleftarrow{u'} & X' \\
 & & \uparrow & & \uparrow \\
 X & \xleftarrow{\quad} & X \times Y' & \xleftarrow{\quad} & X \times X' \\
 u \downarrow & & \downarrow & \swarrow u \times u' & \\
 Y & \xleftarrow{\quad} & Y \times Y' & & .
 \end{array} \tag{1.9.1}$$

Corollary 1.10. *Let $u : T \rightarrow S$ be a squarable morphism, and let $S' \rightarrow S$ be a base-change morphism that is a universal effective epimorphism. Then u is a universal effective epimorphism if and only if*

$$u' : T' = T \times_S S' \longrightarrow S'$$

is one:

$$\begin{array}{ccc}
 T & \xleftarrow{v'} & T' \\
 u \downarrow & & \downarrow u' \\
 S & \xleftarrow{v} & S'.
 \end{array} \tag{1.10.1}$$

Only sufficiency requires proof. If u' is a universal effective epimorphism, then so is vu' , by 1.8. Since $vu' = uv'$, another application of 1.8 shows that u is a universal effective epimorphism.

Remark 1.11. *The same reasoning shows that in 1.10 one may replace “universal effective epimorphism” by “universal epimorphism,” by “universal and effective epimorphism,” or simply by “epimorphism” (and in the last case the hypothesis that u be squarable is plainly unnecessary).*

In the proof of 1.8 we used the following result, which deserves to be stated explicitly.

Proposition 1.12. *Let $u : T \rightarrow S$ be a morphism admitting a section. Then u is an epimorphism; if $T \times_S T$ exists, it is an effective epimorphism; and if, in addition, u is squarable, it is a universal effective epimorphism.*

The first assertion is contained in 1.4(a), and the third follows at once from the second, so it is enough to establish the second. In fact, there is a stronger conclusion: for every functor $F : \mathcal{C}^\circ \rightarrow \mathbf{Set}$, not necessarily representable, the diagram of sets

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact. This may be regarded as a special case of the formalism of Čech cohomology (in dimension 0!), which we merely recall here. Assume only that $T \times_S T$ exists, and set

$$\check{H}^0(T/S, F) = \text{Ker}(F(T) \rightrightarrows F(T \times_S T)).$$

One may plainly regard $\check{H}^0(T/S, F)$ as a contravariant functor of the variable T in $\mathcal{C}_{/S}$: every S -morphism $T' \rightarrow T$ defines a map

$$\check{H}^0(T/S, F) \longrightarrow \check{H}^0(T'/S, F). \tag{+}$$

Fix T, T' in $\mathcal{C}/S$. A familiar calculation shows that, if there exists an S -morphism $T' \rightarrow T$, the corresponding map $(+)$ is independent of the choice of that morphism.⁶

Thus $\check{H}^0(T/S, F)$ may be regarded as a functor on the category associated with the set $\text{Ob}(\mathcal{C}/S)$, preordered by the relation of domination: T' dominates T if there exists an S -morphism $T' \rightarrow T$. In particular, if T and T' are isomorphic in this latter category, that is, if each dominates the other, then $(+)$ is an isomorphism of sets. This applies in particular when T' is the final object of $\mathcal{C}/S$, essentially S itself. The object T always dominates $T' = S$, while the converse holds precisely when T/S has a section. This proves 1.12 in the stronger form announced.

Remark 1.13. *For various applications, the notions introduced and the results stated in this exposé must be developed more generally for a family of morphisms $u_i : T_i \rightarrow S$ with the same target, rather than for a single morphism $u : T \rightarrow S$. Such a family is called epimorphic if, for every object X of $\mathcal{C}$, the corresponding map*

$$X(S) \longrightarrow \prod_i X(T_i)$$

is injective. One similarly introduces the notion of an effective epimorphic family and the universal variants of these notions. Whenever needed below, we shall take for granted that the results of this exposé extend to this more general situation.

Remark 1.14. *Every morphism that is both a monomorphism and an effective epimorphism is an isomorphism.*

Indeed, in the notation of 1.3, the fact that $T \rightarrow S$ is a monomorphism implies that the two morphisms

$$\text{pr}_1, \text{pr}_2 : T \times_S T \rightrightarrows T$$

are equal (and are isomorphisms). But a diagram

$$C \begin{array}{c} \xrightarrow{v} \\ \xrightarrow{v} \end{array} B \xrightarrow{u} A$$

is exact only if u is an isomorphism, as follows immediately from the definition.⁷

2. Descent Morphisms

Recall the following definitions.

Definition 2.1. *Let $f : S' \rightarrow S$ be a morphism such that $S'' = S' \times_S S'$ exists, and let $u' : X' \rightarrow S'$ be an object over S' . A gluing datum on X'/S' , relative to f , is an S'' -isomorphism*

$$c : X''_1 \xrightarrow{\sim} X''_2,$$

⁶Editor's note: this is the following argument, communicated by M. Demazure. Let $f, g : T' \rightarrow T$, and let $\phi : T' \rightarrow T \times_S T$ be the morphism with components f and g , so that $p_1 \circ \phi = f$ and $p_2 \circ \phi = g$. Then $F(\phi) : F(T \times_S T) \rightarrow F(T')$ satisfies $F(\phi) \circ F(p_1) = F(f)$ and $F(\phi) \circ F(p_2) = F(g)$. For every $x \in \check{H}^0(T/S, F)$, however, one has $F(p_1)(x) = F(p_2)(x)$. Applying $F(\phi)$ to both sides gives $F(f)(x) = F(g)(x)$, which shows that f and g induce the same map.

⁷Editor's note: a monomorphism that is an epimorphism need not be an isomorphism. For example, in $\mathcal{C} = \mathbf{Sch}$, the morphism

$$\text{Spec}(\mathbf{F}_p) \amalg \text{Spec}(\mathbf{Z}[1/p]) \longrightarrow \text{Spec}(\mathbf{Z})$$

is a monomorphism and a surjective epimorphism, but is not an isomorphism.

where X_i'' , for $i = 1, 2$, denotes the inverse image, assumed to exist, of X'/S' under the projection $\text{pr}_i : S'' \rightarrow S'$. The gluing datum c is called a descent datum if it satisfies the “cocycle condition”

$$\text{pr}_{3,1}^*(c) = \text{pr}_{3,2}^*(c) \text{pr}_{2,1}^*(c),$$

where $\text{pr}_{i,j}$, for $1 \leq j < i \leq 3$, are the canonical projections from

$$S''' = S' \times_S S' \times_S S'$$

to S'' (and we now assume that S''' also exists), where $\text{pr}_{i,j}^*(c)$ is the inverse image of c , viewed as an S''' -morphism from X_j''' to X_i''' , and where, for each integer k from 1 to 3, X_k''' denotes the inverse image, assumed to exist, of X'/S' under the k -th projection $q_k : S''' \rightarrow S'$.

In the second part of the definition we have therefore used customary identifications and abuses of notation,⁸ which experience shows to be harmless, but which must plainly be avoided in a rigorous exposition of descent theory—a theory that must itself justify, to some extent, these customary abuses of language. Such a formal exposition ([D]) was written by Giraud, with a view to justifying and making precise SGA 1, Exposé VII, which was never written. For a detailed account of the faithfully flat descent results used constantly in this Seminar, one may consult SGA 1, Exposé VIII.

Continue to let $f : S' \rightarrow S$ be a morphism for which $S'' = S' \times_S S'$ exists, and let X be an object over S such that $X' = X \times_S S'$ and $X'' = X \times_S S''$ exist. The inverse images of X' under pr_i , for $i = 1, 2$, then exist and are canonically isomorphic; consequently, X'/S' carries a canonical gluing datum relative to f . When S''' and $X''' = X \times_S S'''$ exist, this is in fact a descent datum. If Y is another object over S satisfying the same conditions as X/S , then for every S -morphism $X \rightarrow Y$, the corresponding S' -morphism $X' \rightarrow Y'$ is compatible with the canonical gluing data on X' and Y' . In particular, if $S' \rightarrow S$ is squarable, then

$$X \mapsto X' = X \times_S S'$$

is a functor from $\mathcal{C}_S$ to the “category of objects over S' equipped with a descent datum relative to f ”—a category whose definition is left to the reader and which is a full subcategory of the “category of objects over S' equipped with a gluing datum relative to f .”

With this understood, we make the following definition.

Definition 2.2. A morphism $f : S' \rightarrow S$ is called a descent morphism (respectively, an effective descent morphism) if f is squarable (that is, $X \times_S S'$ exists for every $X \rightarrow S$) and the preceding functor

$$X \mapsto X' = X \times_S S'$$

from the category $\mathcal{C}_S$ of objects over S to the category of objects over S' equipped with a descent datum relative to f is fully faithful (respectively, an equivalence of categories).

The first of these notions can be expressed using gluing data alone, and therefore without invoking the triple fiber product S''' : f is a descent morphism precisely when it is squarable and $X \mapsto X'$ is a fully faithful functor from $\mathcal{C}_S$ to the category of objects over S' equipped with a gluing datum relative to f . Written out, this means that for any two objects X, Y over S , the following diagram of sets is exact:

$$\text{Hom}_S(X, Y) \longrightarrow \text{Hom}_{S'}(X', Y') \xrightleftharpoons[p_2]{p_1} \text{Hom}_{S''}(X'', Y''). \quad (\text{x})$$

⁸Editor's note: for example, we have identified, on the one hand, $\text{pr}_{2,1}^*(X_1'') = \text{pr}_{3,1}^*(X_1'')$, and, on the other hand, $\text{pr}_{2,1}^*(X_2'') = \text{pr}_{3,2}^*(X_2'')$.

Here $p_i(u')$ denotes the inverse image of $u' \in \text{Hom}_{S'}(X', Y')$ under the projection $\text{pr}_i : S'' \rightarrow S'$, for $i = 1, 2$. Indeed, by definition, the kernel of the pair (p_1, p_2) is precisely the set of S' -morphisms $X' \rightarrow Y'$ compatible with the canonical gluing data.

By the definition of the inverse images Y' and Y'' , there are canonical bijections

$$\text{Hom}_{S'}(X', Y') \simeq \text{Hom}_S(X', Y), \quad \text{Hom}_{S''}(X'', Y'') \simeq \text{Hom}_S(X'', Y).$$

Thus exactness of (x) is equivalent to exactness of

$$\text{Hom}_S(X, Y) \longrightarrow \text{Hom}_S(X', Y) \rightrightarrows \text{Hom}_S(X'', Y), \quad (\text{xx})$$

which is obtained by applying $\text{Hom}_S(-, Y)$ to the diagram in $\mathcal{C}_{/S}$

$$X'' \rightrightarrows X' \longrightarrow X, \quad (\text{xxx})$$

itself obtained from

$$S'' \rightrightarrows S' \longrightarrow S$$

by the base change $X \rightarrow S$. In view of 1.6, this proves the first part of the following proposition.

Proposition 2.3. *Let $f : S' \rightarrow S$ be a morphism. If f is a universal effective epimorphism, then it is a descent morphism. Conversely, every descent morphism is a universal effective epimorphism if S is a squarable object of $\mathcal{C}$, that is, if its product with every object of $\mathcal{C}$ exists.*

It remains to prove the converse. If f is a descent morphism, we must show that it is a universal effective epimorphism: for every morphism $X \rightarrow S$, diagram (xxx) must be exact, or equivalently, its image under $\text{Hom}(-, Z)$ must be an exact diagram of sets for every object Z of $\mathcal{C}$. By hypothesis, $Z \times S$ exists. Let Y be the object of $\mathcal{C}_{/S}$ that it defines. The image of (xxx) under $\text{Hom}(-, Z)$ is then isomorphic to its image under $\text{Hom}_S(-, Y)$, and the latter is exact by the hypothesis on f .

The results concerning descent morphisms may therefore be applied to universal effective epimorphisms. For example:

Proposition 2.4. *Let $f : S' \rightarrow S$ be a descent morphism (for example, a universal effective epimorphism). Then:*

- (a) *For every S -morphism $u : X \rightarrow Y$, the morphism u is an isomorphism (respectively, a monomorphism) if and only if $u' : X' \rightarrow Y'$ is one.*
- (b) *Let X, Y be two subobjects of S , and let X', Y' be their inverse images in S' . Then X is contained in Y (respectively, is equal to Y) if and only if the same is true of X' and Y' .*

For (a), the definition shows that if u' is an isomorphism in the category of objects with gluing datum, then u is an isomorphism. Every isomorphism of objects over S' that is compatible with gluing data is, however, an isomorphism of objects with gluing datum: its inverse is also compatible with the gluing data. For (b), it remains to show that if X' is contained in Y' , that is, if there exists an S' -morphism $X' \rightarrow Y'$, then the same is true of X, Y over S . Since $Y' \rightarrow S'$, and hence $Y'' \rightarrow S''$, is a monomorphism, the morphism $X' \rightarrow Y'$ is automatically compatible with the gluing data and therefore comes from an S -morphism $X \rightarrow Y$.

The proof applies more generally to two objects X, Y over S , with $Y \rightarrow S$ a monomorphism, when one asks whether $X \rightarrow S$ factors through Y : it is enough that $X' \rightarrow S'$ factor through Y' .

Corollary 2.5. *Let $f : S' \rightarrow S$ be a universal effective epimorphism and $g : S \rightarrow T$ a morphism such that $S \times_T S$ exists. Suppose that $S'' = S' \times_S S'$ is also a fiber product of S' with itself over T , that is, that the canonical morphism*

$$S' \times_S S' \xrightarrow{\sim} S' \times_T S'$$

is an isomorphism. Then $g : S \rightarrow T$ is a monomorphism (and the converse is of course true).

Indeed, consider the cartesian diagram

$$\begin{array}{ccc} S' \times_S S' & \xrightarrow{\sim} & S' \times_T S' \\ \downarrow & & \downarrow f \\ S & \longrightarrow & S \times_T S, \end{array} \quad (2.5.1)$$

where the second horizontal arrow is the diagonal morphism. By 1.9, the second vertical arrow f is a universal effective epimorphism; by hypothesis, the first horizontal arrow is an isomorphism. It follows from 2.4(a), or equally from 2.4(b),⁹ that $S \rightarrow S \times_T S$ is also an isomorphism, which says precisely that $g : S \rightarrow T$ is a monomorphism.

Remark 2.6. *The notions introduced in 2.1 in terms of morphisms between certain inverse limits have an evident expression in terms of the contravariant functors defined by the objects S, S', X' under consideration. Subject to the existence of the fiber products considered in 2.1, there is a one-to-one correspondence between gluing data (respectively, descent data) on X'/S' relative to $S' \rightarrow S$, and gluing data (respectively, descent data) for the corresponding objects of*

$$\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}^\circ, \mathbf{Set}).$$

This makes it possible, if desired, to extend these notions to the case in which no existence hypothesis is imposed on fiber products in $\mathcal{C}$.

Remark 2.7. *The notions introduced in this section generalize to families of morphisms. They can also be presented more intrinsically by means of the notion of a sieve (4.1). For these matters the reader is referred to [D].*

3. Universal Effective Equivalence Relations

3.1. Equivalence Relations: Definitions

Definition 3.1.1. *A $\mathcal{C}$ -equivalence relation in an object $X \in \text{Ob } \mathcal{C}$ is a representable subfunctor R of the functor $X \times X$ such that, for every $S \in \text{Ob } \mathcal{C}$, $R(S)$ is the graph of an equivalence relation on $X(S)$.*

This definition applies in particular to the category $\widehat{\mathcal{C}}$. If X is regarded as an object of $\widehat{\mathcal{C}}$, then a $\widehat{\mathcal{C}}$ -equivalence relation in X is simply a subfunctor R of $X \times X$ such that $R(S)$ is the graph of an equivalence relation on $X(S)$, for every object S of $\mathcal{C}$.¹⁰

If R is a $\mathcal{C}$ -equivalence relation in X , let p_i denote the morphism from R to X induced by the projection $\text{pr}_i : X \times X \rightarrow X$. Thus there is a diagram

$$p_1, p_2 : R \rightrightarrows X.$$

⁹Editor's note: applied to $f : S' \times_T S' \rightarrow S \times_T S = Y$.

¹⁰Editor's note: the condition is plainly necessary. Conversely, if $R(S)$ is the graph of an equivalence relation for every $S \in \text{Ob } \mathcal{C}$, this equivalence relation extends to $R(F)$ for every $F \in \text{Ob } \widehat{\mathcal{C}}$: two morphisms $\varphi, \psi : F \rightarrow R$ are declared equivalent if, for every $S \in \text{Ob } \mathcal{C}$ and every $x \in F(S)$, the elements $\varphi(x)$ and $\psi(x)$ are equivalent in $X(S)$.

Definition 3.1.2. A morphism $u : X \rightarrow Z$ is said to be compatible with R if $up_1 = up_2$. An object of $\mathcal{C}$ that is a cokernel of the pair (p_1, p_2) is also called a quotient object of X by R , and is denoted X/R .

Thus one has an exact diagram

$$R \begin{array}{c} \xrightarrow{p_1} \\ \xrightarrow{p_2} \end{array} X \xrightarrow{p} X/R,$$

and X/R represents the covariant functor

$$\mathrm{Hom}_{\mathcal{C}}(X/R, Z) = \{\text{morphisms from } X \text{ to } Z \text{ compatible with } R\}.$$

Since quotient objects are taken in $\mathcal{C}$, the quotient X/R is unique whenever it exists.

These definitions extend at once to a $\widehat{\mathcal{C}}$ -equivalence relation in X . One should note, however, that the identification of the objects of $\mathcal{C}$ with their images in $\widehat{\mathcal{C}}$ does not commute with the formation of quotients: the quotient X/R of X by R in $\mathcal{C}$ is not *a priori* a quotient of X by R in $\widehat{\mathcal{C}}$. One must therefore avoid identifying $\mathcal{C}$ uncritically with its image in $\widehat{\mathcal{C}}$ in questions involving passage to a quotient.¹¹

In what follows, we shall simply say *equivalence relation* to mean a $\widehat{\mathcal{C}}$ -equivalence relation; whenever necessary, we shall specify that a $\mathcal{C}$ -equivalence relation is intended.¹²

Definition 3.1.3. If X is an object of $\mathcal{C}$ over S , an equivalence relation in X over S is an equivalence relation R in X for which the structure morphism $X \rightarrow S$ is compatible with R .

The canonical monomorphism $R \rightarrow X \times X$ then factors through the monomorphism

$$X \times_S X \longrightarrow X \times X$$

and defines an equivalence relation in the object $X \rightarrow S$ of $\mathcal{C}/_S$. Whenever the quotient X/R exists, it is equipped with a canonical morphism to S , and the corresponding object of $\mathcal{C}/_S$ is a quotient of $X \in \mathrm{Ob} \mathcal{C}/_S$ by the preceding equivalence relation. Conversely, if S is a squarable object of $\mathcal{C}$ and $Y \rightarrow S$ is a quotient of X by this equivalence relation in $\mathcal{C}/_S$, then Y is a quotient of X by R in $\mathcal{C}$. In any event, we shall never have to consider quotients in $\mathcal{C}/_S$ that are not already quotients in $\mathcal{C}$.

Definition 3.1.4. Let X (respectively X') be an object of $\mathcal{C}$ equipped with an equivalence relation R (respectively R'). A morphism

$$u : X \longrightarrow X'$$

is said to be compatible with R and R' if the following equivalent conditions hold:

- (i) for every $S \in \mathrm{Ob} \mathcal{C}$, two points of $X(S)$ congruent modulo $R(S)$ are carried by u to two points of $X'(S)$ congruent modulo $R'(S)$;

¹¹Editor's note: here is an illustration that also previews the sequel of this exposé. Let G be a $\mathcal{C}$ -group, let H be a $\mathcal{C}$ -subgroup, and let R be the equivalence relation in G defined by $G \times H \rightarrow G \times G$, $(g, h) \mapsto (g, gh)$ (cf. 3.2). The functor Q defined by $Q(S) = G(S)/H(S)$ is a quotient in $\widehat{\mathcal{C}}$ (by 4.4.9 applied to the coarsest topology; cf. 4.4.2), but it is not in general the quotient one wants. For example, when $\mathcal{C} = \mathbf{Sch}$, there is an exact sequence of affine group schemes

$$1 \longrightarrow \mu_2 \longrightarrow \mathbf{G}_m \xrightarrow{p} \mathbf{G}_m \longrightarrow 1$$

that identifies $\mathbf{G}_m$ with the quotient $\mathbf{G}_m/\mu_2$. Moreover, because p is finite and locally free, $\mathbf{G}_m$ is the sheaf quotient of $\mathbf{G}_m$ by μ_2 in the larger category of sheaves for the fppf topology; cf. 4.6.6(ii) and 6.3.2. By contrast, the quotient Q in $\widehat{\mathcal{C}}$ is not isomorphic to $\mathbf{G}_m$, since, for example, $Q(\mathbf{Z}) = \{1\}$, whereas $\mathbf{G}_m(\mathbf{Z}) = \{\pm 1\}$. Thus Q is not an fppf sheaf and, *a fortiori*, is not representable.

¹²Editor's note: even for a $\widehat{\mathcal{C}}$ -equivalence relation in X , the question of interest is the existence of a quotient in $\mathcal{C}$.

(ii) *there exists a necessarily unique morphism $R \rightarrow R'$ making the diagram*

$$\begin{array}{ccc} R & \longrightarrow & R' \\ \downarrow & & \downarrow \\ X \times X & \xrightarrow{u \times u} & X' \times X' \end{array}$$

commute.

By the universal property of X/R , whenever X/R and X'/R' exist there is a unique morphism v making the diagram

$$\begin{array}{ccc} X & \xrightarrow{p} & X/R \\ u \downarrow & & \downarrow v \\ X' & \xrightarrow{p'} & X'/R' \end{array}$$

commute.

Definition 3.1.5. *A subobject (that is, a representable subfunctor) Y of X is said to be stable under the equivalence relation R if the following equivalent conditions hold:*

- (i) *for every $S \in \text{Ob } \mathcal{C}$, the subset $Y(S)$ of $X(S)$ is stable under $R(S)$;*
- (ii) *the two subobjects of R that are the inverse images of Y under p_1 and p_2 are identical.*

An important special case of a stable subobject of X is the following: the quotient X/R exists and Y is the inverse image in X of a subobject of X/R .

Definition 3.1.6. *Let R be an equivalence relation in X , and let $X' \rightarrow X$ be a morphism. The equivalence relation R' in X' obtained from the cartesian diagram*

$$\begin{array}{ccc} R' & \longrightarrow & R \\ \downarrow & & \downarrow \\ X' \times X' & \longrightarrow & X \times X \end{array}$$

is called the equivalence relation in X' that is the inverse image of the equivalence relation R in X . In particular, if X' is a subobject of X , it is called the equivalence relation induced on X' by R , and is denoted $R_{X'}$.

The morphism $X' \rightarrow X$ is compatible with R' and R ; hence, when the quotients exist, it induces a morphism $X'/R' \rightarrow X/R$ (3.1.4). If X' is a subobject of X , we shall see later that in certain cases one can prove that $X'/R' \rightarrow X/R$ is a monomorphism and therefore identifies X'/R' with a subobject of X/R . In that situation, the inverse image of this subobject in X is a subobject containing X' and stable under R : the *saturation* of X' with respect to the equivalence relation R .

Proposition 3.1.7. *If the subobject Y of X is stable under R , then for $i = 1, 2$ the following two squares are cartesian:*

$$\begin{array}{ccc} R_Y & \longrightarrow & R \\ p_i \downarrow & & \downarrow p_i \\ Y & \longrightarrow & X. \end{array}$$

The proof is immediate.

3.2. Equivalence Relation Defined by a Group Acting Freely

Definition 3.2.1. *Let X be an object of $\mathcal{C}$, and let H be a $\mathcal{C}$ -group acting on X , that is, equipped with a morphism of $\mathcal{C}$ -groups*

$$H \longrightarrow \text{Aut}(X).$$

The group H is said to act freely on X if the following equivalent conditions hold:

- (i) *for every $S \in \text{Ob } \mathcal{C}$, the group $H(S)$ acts freely on $X(S)$;*
- (ii) *the morphism of functors*

$$H \times X \longrightarrow X \times X$$

defined on sets by $(h, x) \mapsto (hx, x)$ is a monomorphism.

In the preceding definition the group H was assumed to act on X on the left. There is of course an analogous notion when the group acts on the right, that is, when one is given a group morphism from the opposite group H° to $\text{Aut}(X)$.

If H acts freely on X , the image of $H \times X$ under the morphism in (ii) is an equivalence relation in X , called the *equivalence relation defined by the action of H on X* . When the quotient of X by this equivalence relation exists, it is denoted $H \backslash X$ (or X/H when H acts on the right). It represents the following covariant functor: for every object Z of $\mathcal{C}$,

$$\text{Hom}(H \backslash X, Z) = \{\text{morphisms from } X \text{ to } Z \text{ invariant under } H\},$$

where a morphism $f : X \rightarrow Z$ is said to be invariant under H if, for every $S \in \text{Ob } \mathcal{C}$, the corresponding morphism $X(S) \rightarrow Z(S)$ is invariant under the group $H(S)$.

Lemma 3.2.2. *Under the assumptions of 3.2.1, let Y be a subobject of X . The following conditions are equivalent:*

- (i) *Y is stable under the equivalence relation defined by H (3.1.5);*
- (ii) *for every $S \in \text{Ob } \mathcal{C}$, the subset $Y(S)$ of $X(S)$ is stable under $H(S)$;*
- (iii) *there exists a necessarily unique morphism f making the diagram*

$$\begin{array}{ccc} H \times Y & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ H \times X & \longrightarrow & X \end{array}$$

commute.

Under these conditions, f defines a morphism of $\widehat{\mathcal{C}}$ -groups

$$H \longrightarrow \text{Aut}(Y),$$

and the equivalence relation in Y defined by this action of H is precisely the equivalence relation induced on Y by the equivalence relation in X defined by the action of H .

The proof is immediate. There is of course an analogous statement for a right action. The action of H on Y defined above will be called the action induced on Y by the given action of H on X .

Consider now two $\mathcal{C}$ -groups H and G , together with a group morphism

$$u : H \longrightarrow G.$$

Then H acts on G by translations (set-theoretically, $hg = u(h)g$), and it acts freely if and only if u is a monomorphism. When it exists, the quotient of G by this action of H is denoted $H \backslash G$. Similarly, one defines a right action of H on G and a quotient G/H . These quotients are functorial with respect to the groups involved. More precisely, one has the following lemma, stated for right quotients.

Lemma 3.2.3. *Let $u : H \rightarrow G$ and $u' : H' \rightarrow G'$ be monomorphisms of $\mathcal{C}$ -groups, and suppose that a morphism of $\mathcal{C}$ -groups*

$$f : G \longrightarrow G'$$

is given. The following conditions are equivalent:

- (i) *f is compatible with the equivalence relations in G and G' defined by H and H' ;*
- (ii) *for every $S \in \text{Ob } \mathcal{C}$, one has $fu(H(S)) \subseteq u'(H'(S))$;*
- (iii) *there exists a necessarily unique multiplicative morphism $g : H \rightarrow H'$ such that the diagram*

$$\begin{array}{ccc} H & \xrightarrow{g} & H' \\ u \downarrow & & \downarrow u' \\ G & \xrightarrow{f} & G' \end{array}$$

commutes.

Under these conditions, if the quotients G/H and G'/H' exist, there is a unique morphism $\bar{f}$ making the diagram

$$\begin{array}{ccc} G & \xrightarrow{f} & G' \\ p \downarrow & & \downarrow p' \\ G/H & \xrightarrow{\bar{f}} & G'/H' \end{array}$$

commute.

The first assertion is proved by reduction to the set-theoretic case. The second follows immediately from (i).

The notions introduced above for general equivalence relations could all be translated into the present setting. We record only the following lemma, whose proof is immediate by reduction to the set-theoretic case.

Lemma 3.2.4. *Let $u : H \rightarrow G$ be a monomorphism of $\mathcal{C}$ -groups, and let G' be a $\mathcal{C}$ -subgroup of G . The subobject G' of G is stable under the equivalence relation defined by H if and only if u factors through the canonical monomorphism $G' \rightarrow G$. Under this condition, the action of H on G' induced by the action on G defined by u is precisely the action arising from the monomorphism $H \rightarrow G'$ through which u factors.*

3.3. Universally Effective Equivalence Relations

Definition 3.3.1. *Let $f : X \rightarrow Y$ be a morphism. The equivalence relation defined by f in X , denoted $\mathcal{R}(f)$, is the $\widehat{\mathcal{C}}$ -equivalence relation in X that is the image of the canonical monomorphism*

$$X \times_Y X \longrightarrow X \times X.$$

Definition 3.3.2. *An equivalence relation R in X is said to be effective if:*

- (i) *R is representable, that is, it is a $\mathcal{C}$ -equivalence relation;*

(ii) the quotient $Y = X/R$ exists in $\mathcal{C}$; ¹³

(iii) the diagram

$$\begin{array}{ccc} R & \xrightarrow[p_2]{p_1} & X \xrightarrow{p} Y \end{array}$$

makes R the fiber square of X over Y , that is, R is the equivalence relation defined by p .

Scholium 3.3.2.1. ¹⁴ If R is an effective equivalence relation in X , then p is an effective epimorphism (1.3). If $f : X \rightarrow Y$ is an effective epimorphism, then $\mathcal{R}(f)$ is an effective equivalence relation in X , one quotient of which is Y . Thus there is a one-to-one “Galois” correspondence between effective equivalence relations in X and effective quotients of X , that is, equivalence classes of effective epimorphisms with source X .

Definition 3.3.3. An equivalence relation R in X is said to be *universally effective* if the quotient $Y = X/R$ exists and, for every $Y' \rightarrow Y$, the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist and R' is the fiber square of X' over Y' . Equivalently, R is effective and $p : X \rightarrow X/R$ is a universal effective epimorphism.

Scholium 3.3.3.1. ¹⁴ As above, there is a one-to-one correspondence between universally effective equivalence relations in X and universal effective quotients of X .

Remark 3.3.3.2. ¹⁴ Suppose that $\mathcal{C}$ is the category of S -schemes, and write $\mathbf{A}_S^1$ for affine one-space over S . Let $R \subset X \times_S X$ be a universally effective equivalence relation, and let $p : X \rightarrow Y$ be its quotient. Then, for every open subset U of Y ,

$$\mathcal{O}(U) = \text{Hom}_S(U, \mathbf{A}_S^1)$$

is the set of elements φ of

$$\mathcal{O}(p^{-1}(U)) = \text{Hom}_S(p^{-1}(U), \mathbf{A}_S^1)$$

such that $\varphi \circ \text{pr}_1 = \varphi \circ \text{pr}_2$. In particular, if R is defined by the action of a group H acting freely on the right on X (cf. 3.2.1), then $\mathcal{O}(U)$ is the set of all $\varphi \in \mathcal{O}(p^{-1}(U))$ such that

$$\varphi(xh) = \varphi(x)$$

for every $S' \rightarrow S$, $x \in X(S')$, and $h \in H(S')$.

Proposition 3.3.4. Let R be a universally effective equivalence relation in X . Let $f : X \rightarrow Z$ be a morphism compatible with R , hence factoring through a morphism $g : X/R \rightarrow Z$. The following conditions are equivalent:

- (i) g is a monomorphism;
- (ii) R is the equivalence relation defined by f .

Indeed, (i) plainly implies (ii), while the converse is precisely 2.5.

Definition 3.3.5. Let H be a $\mathcal{C}$ -group acting freely on X . One says that H acts *effectively*, or that the action of H on X is *effective* (respectively *universally effective*), if the equivalence relation in X defined by the action of H is effective (respectively *universally effective*).

¹³Editor's note: the words “in $\mathcal{C}$ ” have been added.

¹⁴Editor's note: the numbering 3.3.2.1 (respectively 3.3.3.1) has been added for later reference; Remark 3.3.3.2 has also been added.

3.4. (M) -Effectivity

In practice, universal effective epimorphisms are most often difficult to characterize. Nevertheless, one frequently has at hand a number of morphisms of this kind—in scheme theory, for example, faithfully flat quasi-compact morphisms. This leads to the developments below.

3.4.1. We first state several conditions on a family (M) of morphisms of $\mathcal{C}$:

- (a) (M) is stable under base change: every $u : T \rightarrow S$ belonging to (M) is squarable (cf. 1.4.0), and for every $S' \rightarrow S$, the base change

$$u' : T \times_S S' \longrightarrow S'$$

belongs to (M) ;

- (b) the composite of two elements of (M) belongs to (M) ;
 (c) every isomorphism belongs to (M) ;
 (d) every element of (M) is an effective epimorphism.

Conditions (a) and (b) imply:

- (a') the cartesian product of two elements of (M) belongs to (M) . More precisely, let $u : X \rightarrow Y$ and $u' : X' \rightarrow Y'$ be two S -morphisms belonging to (M) . If $Y \times_S Y'$ exists, then $X \times_S X'$ exists and $u \times_S u'$ belongs to (M) .

This follows from the diagram

$$\begin{array}{ccccc}
 & & Y' & \xleftarrow{u'} & X' \\
 & & \uparrow & & \uparrow \\
 X & \longleftarrow & X \times_S Y' & \longleftarrow & X \times_S X' \\
 u \downarrow & & \downarrow & & \swarrow \\
 Y & \longleftarrow & Y \times_S Y' & & u \times_S u'
 \end{array}$$

Likewise, (a) and (d) imply:

- (d') every element of (M) is a universal effective epimorphism.

3.4.2. The family (M_0) of universal effective epimorphisms satisfies conditions (a)–(d) of 3.4.1. Indeed, (a), (c), and (d) hold by definition, and (b) follows from 1.8. Henceforth we assume given a family (M) of morphisms of $\mathcal{C}$ satisfying these conditions; our results apply in particular to the family (M_0) .

Definition 3.4.3. An equivalence relation R in X is said to be *of type (M)* if it is representable and $p_1 \in (M)$; conditions (b) and (c) then imply that $p_2 \in (M)$.

The relation R is said to be *(M) -effective* if it is effective and the canonical morphism $X \rightarrow X/R$ belongs to (M) .

A quotient Y of X is said to be *(M) -effective* if the canonical morphism $X \rightarrow Y$ belongs to (M) .

The following consequences are immediate from the definition.¹⁵

Proposition 3.4.3.1.

¹⁵Editor's note: the numbering 3.4.3.1 has been added for later reference, and the proof of part (i) has been expanded.

- (i) An (M) -effective equivalence relation is of type (M) and is universally effective.
- (ii) An (M) -effective quotient is universally effective (cf. 3.3.3).
- (iii) The maps $R \mapsto X/R$ and $p \mapsto \mathcal{R}(p)$ establish a one-to-one correspondence between (M) -effective equivalence relations in X and (M) -effective quotients of X .
- (iv) To be (M_0) -effective is equivalent to being universally effective.

We prove (i). Since R is (M) -effective, there is a cartesian square

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ p_1 \downarrow & & \downarrow p \\ X & \xrightarrow{p} & X/R, \end{array}$$

and $p \in (M)$. By 3.4.1(a), p_1 and p_2 belong to (M) , so R is of type (M) .

Set $Y = X/R$, and let $Y' \rightarrow Y$ be any morphism. By 3.4.1(a), the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist, and the morphisms $X' \rightarrow Y'$ and $p'_i : R' \rightarrow X'$ belong to (M) . Finally, since $R = X \times_Y X$, associativity of the fiber product gives

$$R' = X \times_Y X \times_Y Y' = X' \times_{Y'} X'.$$

This shows that R' is (M) -effective; in particular, R is universally effective. This proves (i), as well as (iv). Parts (ii) and (iii) follow, taking Definition 3.3.2 into account.

3.4.4. Let H be an S -group whose structure morphism belongs to (M) . If H acts freely on the S -object X , then it defines an equivalence relation of type (M) in X . Indeed, by (a), the fiber product $H \times_S X$ exists and $\text{pr}_2 : H \times_S X \rightarrow X$ belongs to (M) . The action of H is said to be (M) -effective if the equivalence relation in X defined by this action is (M) -effective.

Proposition 3.4.5 ((M)-effectivity and base change). *Let R be an (M) -effective equivalence relation in X over S , and set $Y = X/R$. Let $S' \rightarrow S$ be a base change for which $Y' = Y \times_S S'$ exists. Then $X' = X \times_S S'$ exists, $R' = R \times_S S'$ is an (M) -effective equivalence relation in X' over S' , and*

$$X'/R' \simeq (X/R)'.$$

Indeed, the canonical morphisms $X \rightarrow Y$ and $R \rightarrow Y$ belong to (M) ; hence, by (a'), X' and R' are representable. By associativity of the fiber product, R' is the equivalence relation in X' defined by the canonical morphism $X' \rightarrow Y'$, which belongs to (M) . This proves the assertion.

Proposition 3.4.6 ((M)-effectivity and cartesian products). *Let R (respectively R') be an (M) -effective equivalence relation in X (respectively X') over S . If $(X/R) \times_S (X'/R')$ exists, then $X \times_S X'$ exists, $R \times_S R'$ is an (M) -effective equivalence relation in $X \times_S X'$ over S , and*

$$(X \times_S X')/(R \times_S R') \simeq (X/R) \times_S (X'/R').$$

Set $Y = X/R$ and $Y' = X'/R'$. By (a'), the fiber product $X \times_S X'$ exists and the canonical morphism

$$q : X \times_S X' \longrightarrow Y \times_S Y'$$

belongs to (M) . The formula

$$(X \times_S X') \times_{(Y \times_S Y')} (X \times_S X') \simeq (X \times_Y X) \times_S (X' \times_{Y'} X')$$

(all unlabelled fiber products being taken over S) shows that $R \times_S R'$ is the equivalence relation in $X \times_S X'$ defined by q , which completes the proof.

3.4.7.¹⁶ Suppose that $\mathcal{C}$ has a final object e , and let $f : G \rightarrow G'$ be a morphism of $\mathcal{C}$ -groups belonging to (M) . By 3.4.1(a), the kernel $\text{Ker}(f)$ is represented by $e \times_{G'} G$, and the morphism $\text{Ker}(f) \rightarrow e$ belongs to (M) .

On the other hand, the equivalence relation defined by f is the same as that defined by the action of $\text{Ker}(f)$, say on the right, on G : it is the image of the morphism

$$G \times \text{Ker}(f) \longrightarrow G \times G$$

defined on sets by $(g, h) \mapsto (g, gh)$. Consequently, Scholium 3.3.2.1 gives the following corollary.

Corollary 3.4.7.1. *Suppose that $\mathcal{C}$ has a final object e , and let $f : G \rightarrow G'$ be a morphism of $\mathcal{C}$ -groups belonging to (M) . Then the action of $\text{Ker}(f)$ on G is (M) -effective, and G' is the quotient $G/\text{Ker}(f)$.*

3.5. Construction of Quotients by Descent

It frequently happens that one cannot construct a quotient directly, but can do so after a suitable base change. This subsection gives a useful criterion for that situation.

In 2.1 we defined a descent datum on an object X' over S' , relative to a morphism $S' \rightarrow S$.

Definition 3.5.1. *A descent datum on X' , relative to $S' \rightarrow S$, is said to be effective if X' , equipped with this descent datum, is isomorphic to the inverse image over S' of an object X over S , equipped with its canonical descent datum.*

If $S' \rightarrow S$ is a descent morphism (2.2), then the object X in the definition is unique up to unique isomorphism. To say that $S' \rightarrow S$ is an effective descent morphism (2.2) means that it is a descent morphism and that every descent datum relative to it is effective.

Now consider an equivalence relation R in an object X over S . Let X' , X'' , and X''' be the inverse images of X over

$$S', \quad S'' = S' \times_S S', \quad S''' = S' \times_S S' \times_S S',$$

respectively, and let R' , R'' , and R''' be the equivalence relations obtained from R by inverse image. Suppose that R' in X' is (M) -effective, and consider the quotient $Y' = X'/R'$, an object over S' . By 3.4.5, its two inverse images over S'' are isomorphic to X''/R'' . Thus the S' -object Y' has a canonical gluing datum. Applying in the same way the uniqueness of X'''/R''' , one sees that this is in fact a descent datum. (In this argument it has been tacitly assumed that all the displayed fiber products exist. This holds, in particular, if $S' \rightarrow S$ is squarable, for example if it is a descent morphism.)

Proposition 3.5.2. *Let R be an equivalence relation in the object X over S , and let $S' \rightarrow S$ be a universal effective epimorphism. Assume that every S -morphism whose inverse image over S' belongs to (M) itself belongs to (M) . The following conditions are equivalent:*

- (i) R is (M) -effective in X ;
- (ii) R' is (M) -effective in X' , and the canonical descent datum on X'/R' is effective.

When these conditions hold, the object descended from X'/R' is canonically isomorphic to X/R .

The implication (i) $\Rightarrow$ (ii) is simply 3.4.4 translated into the language of descent. Once the converse is proved, the last assertion of the proposition follows from the fact that a universal

¹⁶Editor's note: this paragraph has been added.

effective epimorphism is a descent morphism (2.3), and hence the descended object is unique up to unique isomorphism.

We prove (ii) $\Rightarrow$ (i). Let $Y' = X'/R'$, and let Y be the descended object. The canonical morphism $p' : X' \rightarrow X'/R' = Y'$ is compatible by construction with the descent data: its two inverse images over S'' coincide with the canonical morphism $X'' \rightarrow X''/R''$. It is therefore the inverse image over S' of an S -morphism $p : X \rightarrow Y$. Since p' belongs to (M) , the assumption on $S' \rightarrow S$ implies that p also belongs to (M) . Moreover, because p' is compatible with the equivalence relation R' , the morphism p is compatible with R , again because a universal effective epimorphism is a descent morphism. Hence there is a morphism

$$R \longrightarrow X \times_Y X.$$

To prove that R is (M) -effective and that Y is isomorphic to X/R , it suffices to prove that this morphism is an isomorphism. After base change from S to S' , it becomes an isomorphism because R' is effective; it is therefore an isomorphism for the same reason as in 2.4.

Notice that the hypothesis in the proposition is satisfied if (M) is the family (M_0) of universal effective epimorphisms and if $\mathcal{C}$ has fiber products (1.10). We obtain the following.

Corollary 3.5.3. *Suppose that $\mathcal{C}$ has fiber products (fiber products over S would suffice). Let R be an equivalence relation in X over S , and let $S' \rightarrow S$ be a universal effective epimorphism. The following conditions are equivalent:*

- (i) R is universally effective in X ;
- (ii) R' is universally effective in X' , and the canonical descent datum on X'/R' is effective.

When these conditions hold, the object descended from X'/R' is canonically isomorphic to X/R .

4. Topologies and Sheaves

The notion of a sieve, and the presentation of the notion of a topology adopted here (4.2.1), which is more intrinsic and in many respects more convenient than the presentation by covering families in [MA], are due to J. Giraud [AS].

4.1. Sieves

Definition 4.1.1. *A sieve of the category $\mathcal{C}$ is a subfunctor C of the final functor*

$$e : \mathcal{C}^\circ \longrightarrow \mathbf{Set}.$$

To every sieve C of $\mathcal{C}$ one associates the set $E(C)$ of objects X of $\mathcal{C}$ such that $C(X) \neq \emptyset$, that is, such that the structural morphism $X \rightarrow e$ factors through C . Thus

$$\begin{cases} X \in E(C) & \Longleftrightarrow & C(X) = e(X) = \{\emptyset\}, \\ X \notin E(C) & \Longleftrightarrow & C(X) = \emptyset. \end{cases} \quad (+)$$

The set $E = E(C)$ has the following property:

$$X \in E \text{ and } \mathrm{Hom}(Y, X) \neq \emptyset \implies Y \in E. \quad (++)$$

(If the set $\text{Ob } \mathcal{C}$ is endowed with its natural preorder, with Y dominating X when there is an arrow from Y to X , then the sets E satisfying $(++)$ are precisely the subsets of $\text{Ob } \mathcal{C}$ that contain every object dominating one of their elements.)¹⁷

Conversely, if E is a subset of $\text{Ob } \mathcal{C}$ having property $(++)$, then E can be written uniquely in the form $E(C)$, with C defined by the formulas $(+)$. Thus sieves of $\mathcal{C}$ correspond bijectively to subsets of $\text{Ob } \mathcal{C}$ satisfying $(++)$. By abuse of language, we shall sometimes call the set $E(C)$ itself a sieve of $\mathcal{C}$.

Let C and C' be two sieves of $\mathcal{C}$.¹⁸ Since both are subfunctors of the final functor e , it is equivalent to say that C dominates C' (for the domination relation in $\text{Ob } \widehat{\mathcal{C}}$), that C is a subfunctor of C' , or that $E(C) \subset E(C')$. In this case we say that C is *finer* than C' . This is an order relation on the set of sieves of $\mathcal{C}$. Moreover,

$$E(C) \cap E(C') = E(C \times C'),$$

so the set of sieves of $\mathcal{C}$ is filtered for the order relation “is finer than.”

Every subset E of $\text{Ob } \widehat{\mathcal{C}}$, and in particular every subset of $\text{Ob } \mathcal{C}$, defines a sieve $C(E)$: the set of $X \in \text{Ob } \mathcal{C}$ such that $F(X) \neq \emptyset$ for at least one $F \in E$ satisfies $(++)$, and hence defines the desired sieve.

This sieve may also be defined as the image of the family of morphisms $\{F \rightarrow e \mid F \in E\}$, in the sense of the following definition.

Definition 4.1.2. *Let $\{F_i \rightarrow F\}$ be a family of morphisms of $\widehat{\mathcal{C}}$ with common target F . The image of this family is the subfunctor of F defined by*

$$S \mapsto \bigcup_i \text{Im } F_i(S) \subset F(S).$$

Proposition 4.1.3. *Formation of the image commutes with base change. In the preceding notation let I be the image of the family $\{F_i \rightarrow F\}$. For every morphism $G \rightarrow F$ in $\widehat{\mathcal{C}}$, the image of the family*

$$\{F_i \times_F G \rightarrow G\}$$

is the subfunctor $I \times_F G$ of G .

Definition 4.1.4.0.¹⁹ *Let C be a sieve of $\mathcal{C}$. If E is a subset of $\text{Ob } \mathcal{C}$ such that $C(E) = C$, then E is called a base of C . Every sieve C has a base, for example $E(C)$.*

We now describe $\text{Hom}(C, F)$, where C is a sieve of $\mathcal{C}$ and $F \in \text{Ob } \widehat{\mathcal{C}}$, by means of a base $\{S_i\}$ of C . For every pair (i, j) there is a diagram in $\widehat{\mathcal{C}}$:

$$\begin{array}{ccc} S_i \times S_j & \longrightarrow & S_i \\ \downarrow & & \downarrow \\ S_j & \longrightarrow & e. \end{array}$$

It gives a diagram of sets

$$\Gamma(F) = \text{Hom}(e, F) \xrightarrow{\sigma} \prod_i \text{Hom}(S_i, F) \xrightarrow[\tau_2]{\tau_1} \prod_{i,j} \text{Hom}(S_i \times S_j, F),$$

¹⁷Editor's note: Here “dominating” is meant in the sense of the preorder just mentioned: Y dominates X if there is an arrow $Y \rightarrow X$. On the other hand, if X, Y are two subobjects of an object Z , one says (cf. 2.4) that Y contains X if $X \subset Y$. To avoid any ambiguity between these usages, the sequel uses “dominating” in the first case and “containing” in the second.

¹⁸Editor's note: The following sentence has been expanded.

¹⁹Editor's note: The numbering 4.1.4.0 has been added to highlight this definition.

with $\tau_1\sigma = \tau_2\sigma$. Hence there is a morphism

$$\mathrm{Hom}(e, F) \longrightarrow \mathrm{Ker} \left(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F) \right).$$

One checks immediately:

Proposition 4.1.4. *There is an isomorphism, functorial in F ,*

$$\mathrm{Hom}(C, F) \xrightarrow{\sim} \mathrm{Ker} \left(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F) \right),$$

for which the diagram

$$\begin{array}{ccc} \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Ker} \left(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F) \right) \\ \parallel & & \uparrow \sim \\ \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Hom}(C, F) \end{array}$$

is commutative; the bottom arrow is induced by the canonical morphism $C \rightarrow e$.

Corollary 4.1.5. *Suppose that the fiber products $S_i \times S_j$ are representable, for example that the S_i are squarable. Then, for every $F \in \mathrm{Ob} \widehat{\mathcal{C}}$, there is an isomorphism*

$$\mathrm{Hom}(C, F) \xrightarrow{\sim} \mathrm{Ker} \left(\prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times S_j) \right).$$

Remark 4.1.6. Let R be a sieve of $\mathcal{C}$, let $\overline{R}$ denote the full subcategory of $\mathcal{C}$ whose object set is $E(R)$, and let

$$i_R : \overline{R} \longrightarrow \mathcal{C}$$

be the inclusion functor. There is an isomorphism, functorial in $F \in \mathrm{Ob} \widehat{\mathcal{C}}$,

$$\mathrm{Hom}(R, F) \xrightarrow{\sim} \Gamma(F \circ i_R),$$

for which the diagram

$$\begin{array}{ccc} \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Hom}(R, F) \\ \downarrow \sim & & \downarrow \sim \\ \Gamma(F) & \longrightarrow & \Gamma(F \circ i_R) \end{array}$$

is commutative; the bottom arrow is induced by i_R .

Definition 4.1.7. *Let $\mathcal{C}$ be a category. A sieve of the object S of $\mathcal{C}$ is a sieve of the category $\mathcal{C}_{/S}$.*

A sieve of S is therefore a subobject of S in $\widehat{\mathcal{C}}$. It corresponds canonically to a subset of $\mathrm{Ob} \mathcal{C}_{/S}$ that contains the source of every arrow whose target it contains. By abuse of language, such a subset will also be called a sieve of S .

4.2. Topologies: Definitions

Definition 4.2.1. A topology on a category $\mathcal{C}$ consists, for every object S of $\mathcal{C}$, of a set $J(S)$ of sieves of S , called *covering sieves* or *refinements* of S , satisfying the following axioms:

- (T 1) For every refinement R of S and every morphism $T \rightarrow S$, the sieve $R \times_S T$ of T is covering (“stability under base change”).
- (T 2) If R, C are two sieves of S , if R is covering, and if for every $T \in \text{Ob } \mathcal{C}$ and every morphism $T \rightarrow R$, the sieve $C \times_S T$ of T is covering, then C is a refinement of S .²⁰
- (T 3) If $C \supset R$ are two sieves of S and R is covering, then C is covering.
- (T 4) For every S , S is a refinement of S .

These axioms may be reformulated as follows. Suppose that a topology $S \mapsto J(S)$ on $\mathcal{C}$ is given. For every $F \in \text{Ob } \widehat{\mathcal{C}}$, let $J(F)$ be the set of subfunctors R of F such that, for every morphism $T \rightarrow F$ in $\widehat{\mathcal{C}}$ with T representable, the sieve $R \times_F T$ of T is covering. By (T 1), this notation agrees with the preceding one. An $R \in J(F)$ is again called a *refinement* of F . The preceding axioms immediately imply:

- (T'0) If $F \supset G$ are objects of $\widehat{\mathcal{C}}$, and if $G \times_F S \in J(S)$ for every $S \in \text{Ob } \mathcal{C}$ and every morphism $S \rightarrow F$, then $G \in J(F)$.
- (T'1) If $G \in J(F)$ and $H \rightarrow F$ is a morphism in $\widehat{\mathcal{C}}$, then $G \times_F H \in J(H)$.
- (T'2) If $F \supset G \supset H$, if $G \in J(F)$, and if $H \in J(G)$, then $H \in J(F)$.
- (T'3) If $F \supset G \supset H$ and $H \in J(F)$, then $G \in J(F)$.
- (T'4) For every $F \in \text{Ob } \widehat{\mathcal{C}}$, one has $F \in J(F)$.

Conversely, if for every $F \in \text{Ob } \widehat{\mathcal{C}}$ one is given a set $J(F)$ of subobjects of F satisfying T'0–T'4, then $S \mapsto J(S)$ defines a topology on $\mathcal{C}$, and these two constructions are inverse to each other.

Properties T'1, T'2, and T'3 imply the further property²¹:

- (T'5) If G and H are two subobjects of F and $G, H \in J(F)$, then $G \cap H \in J(F)$.

Thus $J(F)$, ordered by $\supset$, is filtered. This observation will be useful below.

4.2.2. The topology defined by J is said to be *finer* than the topology defined by J' if $J(S) \supset J'(S)$ for every $S \in \text{Ob } \mathcal{C}$. Equivalently, $J(F) \supset J'(F)$ for every $F \in \text{Ob } \widehat{\mathcal{C}}$.

Every set of topologies on $\mathcal{C}$ has a greatest lower bound. More precisely, let I be an index set and, for each $i \in I$, let $S \mapsto J_i(S)$ be a topology on $\mathcal{C}$. Then

$$J(S) = \bigcap_{i \in I} J_i(S)$$

defines a topology, and it is the greatest lower bound of the given set.

In particular, suppose that for every $S \in \text{Ob } \mathcal{C}$ a set $E(S)$ of sieves of S is given. The *topology generated by these sets* is the least fine topology for which every element of $E(S)$ is a refinement of S .

²⁰Editor's note: In other words, if C is covering “locally with respect to the covering sieve R ,” then C is covering.

²¹Editor's note: T'1 and T'2 suffice: $G \cap H = G \times_F H$ belongs to $J(H)$ by T'1, and hence to $J(F)$ by T'2.

Definition 4.2.3. Let $\{F_i \rightarrow F\}$ be a family of morphisms in $\widehat{\mathcal{C}}$, and let $G \subset F$ be its image (4.1.2). The family is *covering* if $G \in J(F)$. A morphism is *covering* if the one-member family consisting of that morphism is covering.

This definition applies in particular to an inclusion: a sieve C of S is covering if and only if the canonical morphism $C \rightarrow S$ is covering.

Axioms $T'0$ – $T'5$ give the following properties of covering families:

- (C 0) If $\{F_i \rightarrow F\}$ is a family in $\widehat{\mathcal{C}}$ such that every representable base change $\{F_i \times_F S \rightarrow S\}$ is covering, then the original family is covering.
- (C 1) If $\{F_i \rightarrow F\}$ is covering and $G \rightarrow F$ is any morphism, then $\{F_i \times_F G \rightarrow G\}$ is covering (“stability under base change”).
- (C 2) If $\{F_i \rightarrow F\}$ is covering and, for every i , $\{F_{ij} \rightarrow F_i\}$ is covering, then the composite family $\{F_{ij} \rightarrow F\}$ is covering (“stability under composition”).
- (C 3) If $\{G_j \rightarrow F\}$ is covering and $\{F_i \rightarrow F\}$ is a family such that for every j there is an i for which $G_j \rightarrow F$ factors through $F_i \rightarrow F$, then $\{F_i \rightarrow F\}$ is covering (“saturation”).
- (C 4) Every one-member family consisting of an isomorphism is covering.

Notice that (C 2) and (C 3) also imply:

- (C 5) If $\{F_i \rightarrow F\}$ is a family for which there exists a covering family $\{G_j \rightarrow F\}$ such that $\{F_i \times_F G_j \rightarrow G_j\}$ is covering for every j , then $\{F_i \rightarrow F\}$ is covering (“a locally covering family is covering”).

4.2.4. Conversely, let $\mathcal{C}$ be a category having fiber products, and suppose that for every $S \in \text{Ob } \mathcal{C}$ one is given a set of families of morphisms in $\mathcal{C}$ with target S , called covering families, satisfying (C 1)–(C 4), and hence also (C 5). Let $J(S)$ be the set of sieves of S having a covering base (or, equivalently by (C 3), all of whose bases are covering). Then $S \mapsto J(S)$ defines a topology on $\mathcal{C}$, and the two preceding constructions are inverse to one another.

In applications it is inconvenient to consider all covering families, since one often has a fairly simple description of a “sufficient” number of them. This leads to the following definitions.

Definition 4.2.5.0.²² Let $\mathcal{C}$ be a category, and suppose given, for every $S \in \text{Ob } \mathcal{C}$, a set $P(S)$ of families of morphisms in $\mathcal{C}$ with target S . The topology generated by P is the least fine topology for which the given families are covering.

Definition 4.2.5. A pretopology on a category $\mathcal{C}$ consists, for every $S \in \text{Ob } \mathcal{C}$, of a set $R(S)$ of families $\{S_i \rightarrow S\}$, called covering for the pretopology, satisfying:

- (P 1) For every $\{S_i \rightarrow S\} \in R(S)$ and every morphism $T \rightarrow S$, the fiber products $S_i \times_S T$ exist and $\{S_i \times_S T \rightarrow T\} \in R(T)$.
- (P 2) If $\{S_i \rightarrow S\} \in R(S)$ and, for each i , $\{T_{ij} \rightarrow S_i\} \in R(S_i)$, then the composite family $\{T_{ij} \rightarrow S\}$ belongs to $R(S)$.
- (P 3) Every one-member family consisting of an isomorphism is covering.

²²Editor’s note: This definition, which appeared after 4.2.5 in the original, has been placed here because it will be used in 6.2.1 in a slightly more general setting than that of 4.2.5.

Proposition 4.2.6. *For each S , let $J(S)$ be the set of sieves of S that are covering for the topology generated by the pretopology R . Let $J_R(S)$ be the subset of $J(S)$ consisting of the sieves defined by the families in $R(S)$. Then $J_R(S)$ is cofinal in $J(S)$: every refinement of S contains a sieve defined by a family in $R(S)$.*

For every S , let $J'(S)$ be the set of sieves of S containing a sieve in $J_R(S)$. Clearly $J'(S) \subset J(S)$. To prove $J(S) = J'(S)$, it suffices to prove that the $J'(S)$ define a topology, that is, satisfy (T 1)–(T 4). Axioms (T 1), (T 3), and (T 4) are immediate; only (T 2) remains.²³

Let $U \in J'(S)$, let C be a sieve of S , and suppose that $C \times_S T \in J'(T)$ for every $T \rightarrow U$. We must prove $C \in J'(S)$. By definition, U contains a refinement U' defined by a family $\{S_i \rightarrow S\} \in R(S)$. Since (T 3) has already been verified, it is enough to prove $U' \cap C \in J'(S)$; hence we may suppose $U = U'$. By hypothesis, $C \times_S S_i \in J'(S_i)$ for every i . Thus for each i there is a covering family $\{T_{ij} \rightarrow S_i\} \in R(S_i)$ such that $T_{ij} \rightarrow S_i$ factors through $C \times_S S_i \rightarrow S_i$. Consequently $T_{ij} \rightarrow S$ factors through $C \rightarrow S$. Hence C contains the sieve defined by the composite family $\{T_{ij} \rightarrow S\}$, and (P 2) completes the proof.

Axioms (P 1)–(P 3) are those of [MA]. Because pretopologies are so useful in practice, we shall interpret every important result in terms of a pretopology defining the topology under consideration.

Remark 4.2.7. One may introduce a slightly more general notion: for every S , one specifies a set of covering families satisfying (P 1), (P 3), and the conclusion of Proposition 4.2.6. This occurs in particular when the given families satisfy (P 1), (P 3), and (C 5). See [D].

Definition 4.2.8. *Let $\mathcal{C}$ be equipped with a topology, let S be an object of $\mathcal{C}$, and let $P(S')$ be a condition on an argument $S' \in \text{Ob } \mathcal{C}/_S$. Suppose that $\text{Hom}(S'', S') \neq \emptyset$ implies $P(S') \Rightarrow P(S'')$. The condition P is said to hold locally on S for the topology under consideration if the following equivalent conditions hold:*

- (i) *The set of $S' \rightarrow S$ for which $P(S')$ holds is a refinement of S .*
- (ii) *There is a refinement of S such that $P(S')$ holds for every S' in that refinement.*
- (iii) *If the topology is defined by a pretopology, there is a covering family for that pretopology such that $P(S')$ holds for every S' in the family.*

Example 4.2.9. *Let $f : X \rightarrow Y$ be an S -morphism. We say that f is locally an isomorphism if there is a covering family $\{S_i \rightarrow S\}$ such that $f \times_S S_i$ is an isomorphism for every i . Equivalently, there is a refinement R of S such that $X(T) \rightarrow Y(T)$ is an isomorphism for every $T \rightarrow R$.*

Many other examples of this “local” language will appear below.

4.3. Presheaves, Sheaves, and the Sheaf Associated to a Presheaf

Definition 4.3.1. *Let $\mathcal{C}$ be a category. A presheaf of sets on $\mathcal{C}$ is a contravariant functor from $\mathcal{C}$ to the category of sets. The category*

$$\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}^\circ, \mathbf{Set})$$

is called the category of presheaves on $\mathcal{C}$. If $\mathcal{C}$ is equipped with a topology, a presheaf P is separated (respectively, is a sheaf) if, for every $S \in \text{Ob } \mathcal{C}$ and every $R \in J(S)$, the canonical map

$$P(S) = \text{Hom}(S, P) \longrightarrow \text{Hom}(R, P) \quad (+)$$

²³Editor's note: Typographical errors in the original have been corrected in what follows.

is injective (respectively, bijective). The full subcategory of $\widehat{\mathcal{C}}$ whose objects are the sheaves is called the category of sheaves and is denoted by $\mathcal{C}$.²⁴

Proposition 4.3.2. *Let P be a separated presheaf (respectively, a sheaf). For every $H \in \text{Ob } \mathcal{C}$ and every $R \in J(H)$, the canonical map*

$$\text{Hom}(H, P) \longrightarrow \text{Hom}(R, P) \quad (+)$$

is injective (respectively, bijective).

Let P be separated, let H be a presheaf, let $R \in J(H)$, and let $u, v : H \rightarrow P$ satisfy $uj = vj$. For every $f : S \rightarrow H$, with $S \in \text{Ob } \mathcal{C}$, the sieve $R \times_H S$ is a refinement of S , and $ufj_S = vfj_S$. This is expressed by the diagram

$$\begin{array}{ccccc} R \times_H S & \xrightarrow{j_S} & S & & \\ f_R \downarrow & & f \downarrow & & \\ R & \xrightarrow{j} & H & \xrightarrow[u]{v} & P. \end{array}$$

Since P is separated, it follows that $uf = vf$. This holds for every representable S , so $u = v$.

Now suppose that P is a sheaf. Let $g : R \rightarrow P$; we show that it factors through H . For every $f : S \rightarrow H$, with $S \in \text{Ob } \mathcal{C}$, the morphism $g \circ f_R : R \times_H S \rightarrow P$ factors uniquely through S , thereby defining a morphism $h : S \rightarrow P$. By uniqueness, h is plainly functorial in f :

$$\begin{array}{ccccc} R \times_H S & \xrightarrow{f_R} & R & \xrightarrow{g} & P \\ j_S \downarrow & & j \downarrow & & h \uparrow \\ S & \xrightarrow{f} & H & \dashrightarrow & P. \end{array}$$

Thus, for every S , we have defined a map $H(S) \rightarrow P(S)$ functorial in S , and hence a morphism $H \rightarrow P$ with the required properties.

Corollary 4.3.2.1.²⁵ *Let R and F be two sheaves. If R is a refinement of F , then $R = F$.*

Indeed, suppose that R is a refinement of F , and write $j : R \hookrightarrow F$ for the inclusion. By 4.3.2, $\text{Hom}(F, R) = \text{End}(R)$; hence there is a $\pi : F \rightarrow R$ such that $\pi j = \text{id}_R$. Likewise, $\text{End}(F) = \text{Hom}(R, F)$, and $j\pi j = j$ implies $j\pi = \text{id}_F$. Thus j is an isomorphism.

Proposition 4.3.3 ([AS], 1.3). *Let $\mathcal{C}$ be a category and P a presheaf on $\mathcal{C}$. For every $S \in \text{Ob } \mathcal{C}$, let $J(S)$ be the set of sieves R of S such that, for every $T \rightarrow S$, the map*

$$\text{Hom}(T, P) \longrightarrow \text{Hom}(R \times_S T, P) \quad (+)$$

is injective (respectively, bijective). Then the $J(S)$ define a topology on $\mathcal{C}$; that is, they satisfy (T 1)–(T 4).

Corollary 4.3.4. *For every $S \in \text{Ob } \mathcal{C}$, let $K(S)$ be a family of sieves satisfying (T 1), and let P be a presheaf on $\mathcal{C}$. Then P is separated (respectively, a sheaf) for the topology generated by the $K(S)$ if and only if, for every $S \in \text{Ob } \mathcal{C}$ and every $R \in K(S)$, the canonical map*

$$\text{Hom}(S, P) \longrightarrow \text{Hom}(R, P) \quad (+)$$

is injective (respectively, bijective).

²⁴Editor's note: If Q is a subpresheaf of a separated presheaf P , then Q is separated. Indeed, for every sieve R of S , the composite $Q(S) \hookrightarrow P(S) \hookrightarrow P(R)$ is injective and factors through $Q(S) \rightarrow Q(R)$.

²⁵Editor's note: This corollary has been added.

Corollary 4.3.5. *For each $S \in \text{Ob } \mathcal{C}$, let $R(S)$ be a set of families of morphisms in $\mathcal{C}$ with target S , satisfying (P 1), for example a pretopology. Let P be a presheaf on $\mathcal{C}$. Then P is separated (respectively, a sheaf) for the topology generated by R if and only if, for every $S \in \text{Ob } \mathcal{C}$ and every family $\{S_i \rightarrow S\} \in R(S)$, the map*

$$P(S) \longrightarrow \prod_i P(S_i)$$

is injective (respectively, the diagram

$$P(S) \longrightarrow \prod_i P(S_i) \rightrightarrows \prod_{i,j} P(S_i \times_S S_j)$$

is exact).

Definition 4.3.6. *The canonical topology on a category $\mathcal{C}$ is the finest topology for which every representable functor is a sheaf.*

Corollary 4.3.7. *A sieve R of S is a refinement for the canonical topology if and only if, for every morphism $T \rightarrow S$ in $\mathcal{C}$ and every $X \in \text{Ob } \mathcal{C}$, the canonical map*

$$\text{Hom}(T, X) \longrightarrow \text{Hom}(R \times_S T, X)$$

is bijective.

Definition 4.3.8. *A sieve covering for the canonical topology is called a universal effective epimorphic sieve.*

Corollary 4.3.9. *A universal effective epimorphic family defines a universal effective epimorphic sieve. Conversely, every squarable family defining a universal effective epimorphic sieve is universal effective epimorphic.*

Return now to a category $\mathcal{C}$ equipped with an arbitrary topology and to the construction of the sheaf associated to a presheaf P . Let S be an object of $\mathcal{C}$. If $R \supset R'$ are two refinements of S , there is a diagram

$$\begin{array}{ccc} \text{Hom}(S, P) & \longrightarrow & \text{Hom}(R, P) \\ & \searrow & \downarrow \\ & & \text{Hom}(R', P). \end{array}$$

As already observed, the ordered set $J(S)$ is filtered. Since $S \in J(S)$, there is an evident morphism

$$\text{Hom}(S, P) \longrightarrow \varinjlim_{R \in J(S)} \text{Hom}(R, P).$$

Definition 4.3.10.0.²⁶ Set

$$\check{H}^0(S, P) = \varinjlim_{R \in J(S)} \text{Hom}(R, P).$$

One checks that $\check{H}^0(S, P)$ depends functorially on S , and hence defines a functor LP by

$$\text{Hom}(S, LP) = \check{H}^0(S, P) = \varinjlim_{R \in J(S)} \text{Hom}(R, P). \quad (++)$$

²⁶Editor's note: The numbering 4.3.10.0 has been added for later reference. Moreover, it follows from the definition that if $Q \rightarrow P$ is a monomorphism, then the same is true of $LP \rightarrow LQ$; thus L “preserves monomorphisms” (see also 4.3.16 for the more general statement that L “commutes with finite inverse limits”).

By construction there are morphisms

$$\ell_P : P \longrightarrow LP, \quad z_R : \text{Hom}(R, P) \longrightarrow \text{Hom}(S, LP).$$

Lemma 4.3.10.

(i) *For every refinement R of S and every $u : R \rightarrow P$, the diagram*

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ u \uparrow & & \uparrow z_R(u) \\ R & \xrightarrow{i_R} & S \end{array}$$

is commutative.

- (ii) *For every morphism $v : S \rightarrow LP$, there exist a refinement R of S and a morphism $u : R \rightarrow P$ such that $v = z_R(u)$.*
- (iii) *Let Q be a functor and let $u, v : Q \rightarrow P$ satisfy $\ell_P u = \ell_P v$. Then the kernel of the pair (u, v) is a refinement of Q .*
- (iv) *Let $u : R \rightarrow P$ and $u' : R' \rightarrow P$. The equality $z_R(u) = z_{R'}(u')$ holds if and only if there is a refinement $R'' \subset R \cap R'$ of S on which u and u' coincide.*

Proof. For (i), we must prove $z_R(u)i_R = \ell_P u$. It is enough to compare the composites of these two morphisms with every morphism $g : T \rightarrow R$, where T is representable. Put $f = i_R g$ and $R' = R \times_S T$.²⁷ We have

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ u \uparrow & & \uparrow z_R(u) \\ R & \xrightarrow{i_R} & S \\ p \uparrow & \swarrow g & \uparrow f \\ R' & \xrightarrow[\sim]{i_{R'}} & T. \end{array}$$

By the definition of ℓ_P , $\ell_P u g = z_R(u)f$; this is the special case of the assertion in which i_R is an isomorphism. But $z_R(u)f = z_R(u)i_R g$, proving (i).

Assertions (ii) and (iv) merely express the definition of $\text{Hom}(S, LP)$ as a direct limit.

For (iii), let K be the kernel of (u, v) . For every morphism $f : S \rightarrow Q$, with S representable, $K \times_Q S$ is a subfunctor of the kernel of $uf, vf : S \rightarrow P$. By $T'0$, it is therefore enough to prove the assertion when $Q = S$ is representable. In that case, (ii) and (iv) show that K contains a refinement of S , and hence is itself a refinement of S .

One finally checks that $P \mapsto LP$ defines a functor

$$L : \widehat{\mathcal{C}} \longrightarrow \widehat{\mathcal{C}},$$

and that $P \mapsto \ell_P$ defines a morphism of functors

$$\ell : \text{Id}_{\widehat{\mathcal{C}}} \longrightarrow L.$$

²⁷Editor's note: Let $p : R' \rightarrow R$ be the projection. Since i_R is a monomorphism, $p = gi_{R'}$. Let g' be the section of $i_{R'}$ defined by g . Then $pg' = g$, hence $pg'i_{R'} = gi_{R'} = p$. This implies $g'i_{R'} = \text{id}_{R'}$, so $i_{R'} : R' \rightarrow T$ is an isomorphism with inverse g' .

We now state the essential result.

Proposition 4.3.11.

- (i) For every presheaf P , the presheaf LP is separated and $\ell_P : P \rightarrow LP$ is covering (4.2.3).
- (ii) If P is a sheaf, then $P \rightarrow LP$ is an isomorphism.
- (iii) For every presheaf P and every separated presheaf (respectively, sheaf) F , the map

$$\mathrm{Hom}(\ell_P, F) : \mathrm{Hom}(LP, F) \longrightarrow \mathrm{Hom}(P, F)$$

is injective (respectively, bijective).

- (iv) If P is separated, then $\ell_P : P \rightarrow LP$ is a covering monomorphism (so P is a refinement of LP), and LP is a sheaf.

Proof. (i) First, $P \rightarrow LP$ is covering. Indeed, for every $v : S \rightarrow LP$, Lemma 4.3.10(i),(ii) supplies a refinement R of S and a commutative diagram (the map from R to the fiber product is induced by $v' : R \rightarrow P$ and $i_R : R \rightarrow S$)

$$\begin{array}{ccccc} P \times_{LP} S & \longrightarrow & P & & R & \longrightarrow & P \times_{LP} S \\ \downarrow & & \downarrow \ell_P & & \searrow i_R & & \downarrow \\ S & \xrightarrow{v} & LP & & & & S, \end{array}$$

so $P \times_{LP} S \rightarrow S$ is covering. By (C 0), $P \rightarrow LP$ is covering.

Now suppose that two morphisms $v_1, v_2 : S \rightarrow LP$ induce the same morphism on a refinement R of S . There are refinements R_i , $i = 1, 2$, and morphisms $u_i : R_i \rightarrow P$ with $z_{R_i}(u_i) = v_i$. Replacing R by a smaller refinement, we may suppose $R_1 = R_2 = R$. The commutative diagram of Lemma 4.3.10(i) then gives $\ell_P u_1 = \ell_P u_2$. By 4.3.10(iii), u_1 and u_2 coincide on a refinement of R , hence on a refinement of S ; by 4.3.10(iv), $z_R(u_1) = z_R(u_2)$. Thus LP is separated.

(ii) This is clear: if P is a sheaf, then $\mathrm{Hom}(S, P) \rightarrow \mathrm{Hom}(R, P)$ is already an isomorphism for every refinement R of S .

(iii) Let $u, v : LP \rightarrow F$ satisfy $u\ell_P = v\ell_P$. To show $u = v$, it is enough to prove $uf = vf$ for every $f : S \rightarrow LP$ with S representable. There are a refinement R of S and a morphism $g : R \rightarrow P$ with $f = z_R(g)$. The morphisms uf and vf agree on R , where both equal $u\ell_P g = v\ell_P g$. If F is separated, they therefore agree on S , proving injectivity. If F is a sheaf, the commutative diagram

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ \downarrow & & \downarrow \\ F & \xrightarrow{\sim} & LF \end{array}$$

shows that $\mathrm{Hom}(\ell_P, F)$ is also surjective.

(iv) Suppose that P is separated. To show that $P \rightarrow LP$ is a monomorphism, it is enough to consider $u, v : S \rightarrow P$, with S representable, satisfying $\ell_P u = \ell_P v$. By 4.3.10(iii), u and v agree on a refinement of S , and therefore agree because P is separated. Thus $P \rightarrow LP$ is a monomorphism; since it is covering by (i), P is a refinement of LP .

It remains to show that LP is a sheaf. We already know from (i) that it is separated. Let $S \in \mathrm{Ob} \mathcal{C}$, let R be a refinement of S , and let $h : R \rightarrow LP$. We must find $u : S \rightarrow LP$

with $ui_R = h$. Since P is a refinement of LP , $R' = P \times_{LP} R$ is a refinement of R , and hence of S . Put $u = z_{R'}(h')$; the situation is

$$\begin{array}{ccccc} P & \xrightarrow{\ell_P} & LP & & \\ \uparrow h' & & \uparrow h & \nwarrow u & \\ R' & \xrightarrow{j} & R & \xrightarrow{i_R} & S. \end{array}$$

Here the diagonal arrow is $u : S \rightarrow LP$. We have $ui_{R'} = \ell_P h' = hj$, hence $ui_R j = hj$. Since R' is a refinement of R and LP is separated, Proposition 4.3.2 gives $ui_R = h$.

Corollary 4.3.12.²⁸ *Let F be a sheaf and let R be a subobject of F in $\widehat{\mathcal{C}}$. Then R is a separated presheaf, $\ell_R : R \rightarrow LR$ is a covering monomorphism, and there is a commutative diagram*

$$\begin{array}{ccc} R & \xrightarrow{i} & F \\ & \searrow \ell_R & \uparrow j \\ & & LR. \end{array}$$

Consequently, R is a refinement of F if and only if j is an isomorphism.

We already observed that R is separated and that j is a monomorphism; see editor notes (24) and (26). By Proposition 4.3.11(iv), ℓ_R is a covering monomorphism. Thus, if j is an isomorphism, R is a refinement of F . Conversely, if i is covering, then so is j , and Proposition 4.3.2.1 shows that j is an isomorphism.

Remark 4.3.13. If $J'(S)$ is a cofinal subset of $J(S)$, then

$$\mathrm{Hom}(S, LP) = \varinjlim_{R \in J'(S)} \mathrm{Hom}(R, P).$$

In particular, let $S \mapsto R(S)$ be a pretopology generating the given topology. The functor L can be described using the covering families belonging to $R(S)$. Writing out the preceding formula recovers the construction of [MA].

Let $i : \widetilde{\mathcal{C}} \rightarrow \widehat{\mathcal{C}}$ be the inclusion functor. Proposition 4.3.11 gives the following theorem.

Theorem 4.3.14. *There is a unique functor $a : \widehat{\mathcal{C}} \rightarrow \widetilde{\mathcal{C}}$ for which the diagram*

$$\begin{array}{ccc} \widehat{\mathcal{C}} & \xrightarrow{L} & \widehat{\mathcal{C}} \\ \downarrow a & & \downarrow L \\ \widetilde{\mathcal{C}} & \xrightarrow{i} & \widehat{\mathcal{C}} \end{array}$$

is commutative; that is, $L(L(P))$ is a sheaf for every presheaf P . The functors i and a are adjoint: for every presheaf P and every sheaf F , there is an isomorphism, functorial in P and F ,

$$\mathrm{Hom}_{\widehat{\mathcal{C}}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathcal{C}}}(a(P), F),$$

or simply

$$\mathrm{Hom}(P, F) \simeq \mathrm{Hom}(a(P), F).$$

Definition 4.3.15. *The sheaf $a(P)$ is called the sheaf associated to the presheaf P .*

²⁸Editor's note: The statement and proof of this corollary have been expanded.

Remark 4.3.16. Since L is constructed using inverse limits and filtered direct limits, it commutes with finite inverse limits.²⁹

Moreover, under the identification $L(P \times P) = LP \times LP$, the morphism $\ell_{P \times P}$ is identified with $\ell_P \times \ell_P$. Thus, for example, if P is a presheaf of groups, then LP is canonically a presheaf of groups and the canonical morphism $P \rightarrow LP$ is a morphism of groups. The same holds for the functor a . Consequently, if P is a presheaf of groups and F a sheaf of groups, there is an isomorphism

$$\mathrm{Hom}_{\widehat{\mathcal{C}\text{-gr.}}}(P, i(F)) \simeq \mathrm{Hom}_{\widehat{\mathcal{C}\text{-gr.}}}(a(P), F).$$

See [D] for further details.

4.3.17. Let $\mathcal{V}$ be any category. A *presheaf on $\mathcal{C}$ with values in $\mathcal{V}$* is a contravariant functor from $\mathcal{C}$ to $\mathcal{V}$. Before defining sheaves with values in $\mathcal{V}$, we recall the definition of the inverse limit of a functor. Let $\mathcal{R}$ and $\mathcal{V}$ be categories, and let

$$F : \mathcal{R}^\circ \longrightarrow \mathcal{V}$$

be a contravariant functor. Write $\varprojlim F$ for the object of $\widehat{\mathcal{V}}$ defined by

$$\begin{aligned} (\varprojlim F)(X) &= \mathrm{Hom}_{\widehat{\mathcal{V}}}(X, \varprojlim F) \\ &= \varprojlim_{U \in \mathrm{Ob} \mathcal{R}} \mathrm{Hom}_{\mathcal{V}}(F(U), X) \\ &= \mathrm{Hom}(c_X, F), \end{aligned}$$

where X varies over the objects of $\mathcal{V}$, where c_X denotes the contravariant functor from $\mathcal{R}$ to $\mathcal{V}$ that sends every object to X and every arrow to id_X , and where the last Hom is taken in $\mathrm{Hom}(\mathcal{R}^\circ, \mathcal{V})$. If $\mathcal{R}$ has a final object $e_{\mathcal{R}}$, then $\varprojlim F = F(e_{\mathcal{R}})$. If $\mathcal{V} = \mathbf{Set}$, the functor $\varprojlim$ is identified with Γ .

Let S be an object of $\mathcal{C}$ and R a sieve of S . Let $\overline{R}$ be the full subcategory of $\mathcal{C}_{/S}$ whose object set is $E(R)$, and let $i_R : \overline{R} \rightarrow \mathcal{C}_{/S}$ be the canonical functor. A presheaf P on $\mathcal{C}$ with values in $\mathcal{V}$ restricts to a functor

$$P_S : (\mathcal{C}_{/S})^\circ \longrightarrow \mathcal{V}.$$

The functor i_R induces a morphism in $\widehat{\mathcal{V}}$:

$$P(S) = P_S(S) = \varprojlim P_S \longrightarrow \varprojlim (P_S \circ i_R).$$

We write

$$\varprojlim_R P = \varprojlim (P_S \circ i_R)$$

for the latter object of $\widehat{\mathcal{V}}$. In view of 4.1.6, Definition 4.3.1 generalizes as follows.

Definition 4.3.18. A presheaf P on $\mathcal{C}$ with values in $\mathcal{V}$ is separated (respectively, a sheaf) if, for every $S \in \mathrm{Ob} \mathcal{C}$ and every $R \in J(S)$, the canonical morphism in $\widehat{\mathcal{V}}$

$$P(S) \longrightarrow \varprojlim_R P$$

is a monomorphism (respectively, an isomorphism).

²⁹Editor's note: In particular, if K is the kernel of a pair of morphisms of presheaves $u, v : Q \rightrightarrows P$, then LK is the kernel of $Lu, Lv : LQ \rightrightarrows LP$. This will be used in 4.4.5.

When $\mathcal{V}$ is the category **Gr** of groups (or any other category of sets endowed with algebraic structures defined by finite inverse limits), the following notions are equivalent (cf. [D]): a presheaf on $\mathcal{C}$ with values in **Gr** whose underlying presheaf of sets is a sheaf, and a group object in the category of sheaves of sets. With these identifications understood, we shall always regard sheaves with values in a category of sets endowed with algebraic structures defined by finite inverse limits as sheaves of sets equipped, in $\widetilde{\mathcal{C}}$, with the corresponding algebraic structure.

4.4. Exactness Properties of the Category of Sheaves

Theorem 4.4.1.

- (i) *Arbitrary inverse limits exist in $\widetilde{\mathcal{C}}$, and “are computed in $\widehat{\mathcal{C}}$ ”; that is, the inclusion functor*

$$i : \widetilde{\mathcal{C}} \longrightarrow \widehat{\mathcal{C}}$$

commutes with inverse limits. More explicitly, if (X_α) is an inverse system of sheaves, the presheaf

$$\varprojlim_{\alpha} i(X_\alpha) : S \longmapsto \varprojlim_{\alpha} X_\alpha(S)$$

is a sheaf, and

$$i\left(\varprojlim_{\alpha} X_\alpha\right) = \varprojlim_{\alpha} i(X_\alpha).$$

- (ii) *Arbitrary direct limits exist in $\widetilde{\mathcal{C}}$. If (X_α) is a direct system of sheaves, then*

$$\varinjlim_{\alpha} X_\alpha = a\left(\varinjlim_{\alpha} i(X_\alpha)\right),$$

where $\varinjlim_{\alpha} i(X_\alpha)$ is the presheaf direct limit of the $i(X_\alpha)$:

$$\varinjlim_{\alpha} i(X_\alpha) : S \longmapsto \varinjlim_{\alpha} X_\alpha(S).$$

- (iii) *The functor*

$$a : \widehat{\mathcal{C}} \longrightarrow \widetilde{\mathcal{C}}$$

commutes with arbitrary direct limits and finite inverse limits.

Assertions (i) and (ii) follow formally from the adjunction formula (4.3.14), and³⁰ the first assertion of (iii) follows from (ii). The second assertion of (iii) was already noted in 4.3.16.

Scholium 4.4.2. *This theorem makes it possible to use the following method for proving in $\widetilde{\mathcal{C}}$ a statement involving both arbitrary direct limits and finite inverse limits (for example, “every epimorphism is universally effective”; cf. below). One begins by proving the corresponding statement in the category of sets, then extends it pointwise to the category of presheaves. One then uses the preceding theorem to pass from the category of presheaves to the category of sheaves. Many examples of this method will occur below (4.4.3, 4.4.6, 4.4.9, and so forth).*

³⁰Editor’s note: the original has been modified here.

Finally, observe that statements concerning the category of presheaves are formally corollaries of the corresponding statements concerning the category of sheaves. It is enough to take the coarsest (“chaotic”) topology, namely the topology defined by

$$J(S) = \{S\} \quad \text{for every } S \in \text{Ob } \mathcal{C};$$

every functor is a sheaf for this topology.

Proposition 4.4.3. *Let $\mathcal{F} = \{F_i \rightarrow F\}$ be a family of morphisms of sheaves. The following conditions are equivalent:*

- (i) $\mathcal{F}$ is an epimorphic family;
- (ii) $\mathcal{F}$ is a universal effective epimorphic family (1.13);
- (iii) $\mathcal{F}$ is covering (4.2.3);
- (iv) the sheaf image of $\mathcal{F}$, that is, the sheaf associated with the presheaf image of $\mathcal{F}$ (4.1.2), is F .

The equivalence of (iii) and (iv) follows from 4.3.12. The remaining equivalences will follow from the next lemmas.

Lemma 4.4.4. *Let $f : X \rightarrow Y$ be a monomorphism of sheaves that is also an epimorphism. Then f is an isomorphism.*

The lemma is clear first of all in the category of sets. Let us next prove it in the category of presheaves. Consider the presheaf

$$V : S \mapsto Y(S) \amalg_{X(S)} Y(S);$$

this is the amalgamated sum of Y and Y over X in the category of presheaves.³¹ Let $i_1, i_2 : Y \rightrightarrows V$ be the two coordinate morphisms. If $X \rightarrow Y$ is an epimorphism in the category of presheaves, then $i_1 = i_2$. In that case, for every S , the map $X(S) \rightarrow Y(S)$ is surjective; since it is also injective, it is bijective, and hence $X \rightarrow Y$ is an isomorphism.

Finally, work in the category $\mathcal{C}$ of sheaves. By 4.4.1(ii), the amalgamated sum in $\mathcal{C}$ of the sheaves Y and Y over X is the sheaf Z associated with the presheaf V . Consider the diagram

$$X \xrightarrow{f} Y \xrightleftharpoons[i_2]{i_1} V \xrightarrow{\tau} Z = a(V).$$

We have $i_1 f = i_2 f$, hence $\tau i_1 f = \tau i_2 f$, and therefore $\tau i_1 = \tau i_2$, since f is an epimorphism in $\mathcal{C}$. By part (iii) of the following lemma, the presheaf V is separated, so τ is a monomorphism (4.3.11(iv)). Thus $i_1 = i_2$, and we saw above that this implies that $f : X \rightarrow Y$ is an isomorphism. This proves Lemma 4.4.4, once it has been verified that V is separated.

Let $R_\emptyset$ denote the presheaf that assigns the empty set to every $S \in \text{Ob } \mathcal{C}$; it is an initial object of $\widehat{\mathcal{C}}$.³²

Lemma 4.4.5.

³¹Editor's note: the remainder of the proof has been slightly modified.

³²Editor's note: part (i) of Lemma 4.4.5 has been corrected, and the proofs of all three parts have been given in detail.

- (i) Suppose that $R_\emptyset \notin J(S)$ for every $S \in \text{Ob } \mathcal{C}$. If (X_i) is a family of separated presheaves, then the coproduct presheaf $\coprod_i X_i$ is separated.³³
- (ii) Consider an equivalence relation in the category of presheaves

$$X \begin{smallmatrix} \xrightarrow{u} \\ \xrightarrow{v} \end{smallmatrix} Y,$$

and let $w : Y \rightarrow Z$ be the quotient. If X is a sheaf and Y is separated, then Z is separated.

- (iii) Consider an amalgamated sum in the category of presheaves, in which u and u' are monomorphisms:

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ u' \downarrow & & \downarrow \\ Y' & \longrightarrow & V. \end{array}$$

If Y and Y' are separated and X is a sheaf, then V is separated.

Proof. For (i), set $X = \coprod_i X_i$. Let $S \in \text{Ob } \mathcal{C}$, let $\tau : R \hookrightarrow S$ be a refinement of S , and let $x_1, x_2 \in X(S)$ satisfy $x_1 \circ \tau = x_2 \circ \tau$. There are indices i, j such that $x_1 \in X_i(S)$ and $x_2 \in X_j(S)$. Since $R \neq R_\emptyset$, there is a morphism $\phi : T \rightarrow R$ with $T \in \text{Ob } \mathcal{C}$. Then

$$x_1 \circ \tau \circ \phi = x_2 \circ \tau \circ \phi.$$

Since $X(T)$ is the disjoint union of the $X_k(T)$, it follows that $i = j$. Because X_i is separated and is a subobject of X , the map

$$X_i(S) \longrightarrow X_i(R) \longrightarrow X(R)$$

is injective. Hence $x_1 = x_2$, which proves that X is separated.

We next prove (iii). Let $i : Y \rightarrow V$ and $j : Y' \rightarrow V$ be the canonical morphisms, and let K be the kernel of the pair

$$Y \times Y' \begin{smallmatrix} \xrightarrow{p} \\ \xrightarrow{q} \end{smallmatrix} V, \quad p = i \circ \text{pr}_1, \quad q = j \circ \text{pr}_2.$$

For $S \in \text{Ob } \mathcal{C}$, the monomorphisms u, u' allow us to identify $X(S)$ with its image in $Y(S)$, respectively in $Y'(S)$. Write $Z(S)$, respectively $Z'(S)$, for the complement. Then $V(S)$ identifies with the disjoint union of $Z(S)$, $Z'(S)$, and $X(S)$. It follows at once that

$$i(S) : Y(S) \longrightarrow V(S), \quad j(S) : Y'(S) \longrightarrow V(S)$$

are injective and that

$$(u \times u')(S) : X(S) \longrightarrow K(S)$$

is bijective. Consequently i and j are monomorphisms, and $u \times u' : X \rightarrow K$ is an isomorphism.

³³Editor's note: in general, the coproduct of two sheaves F, G is not a sheaf. Indeed, let $S_1, S_2 \in \text{Ob } \mathcal{C}$. Suppose that the coproduct $S = S_1 \coprod S_2$ exists in $\mathcal{C}$, and that the fiber product $S_1 \times_S S_2$ is an initial object $\emptyset$ of $\mathcal{C}$ (cf. I, 1.8). Let R be the sieve of S with basis $\{S_1, S_2\}$. Then $(F \coprod G)(R)$ is the disjoint union of $F(S) \coprod G(S)$ and the sets $F(S_i) \times G(S_j)$, for $i \neq j$; thus $F \coprod G$ is not a sheaf in general. On the other hand, let $\mathcal{C}$ be the category with a single object S and id_S as its only morphism, equipped with the topology defined by $J(S) = \{R_\emptyset, S\}$. Then the only separated presheaves are $R_\emptyset$ and $X = \mathbf{h}_S$ (which is a sheaf), and $X \coprod X$ is not separated.

Set $U = Y \amalg Y'$. For a refinement $\tau : R \hookrightarrow S$, there is a commutative diagram

$$\begin{array}{ccc} U(S) & \xrightarrow{U(\tau)} & U(R) \\ \downarrow & & \downarrow \\ V(S) & \xrightarrow{V(\tau)} & V(R). \end{array}$$

Let $v_1, v_2 \in V(S)$ have the same image in $V(R)$. By the definition of V , they lift to elements $y_1, y_2 \in U(S)$; let z_1, z_2 denote their images in $U(R)$. Then z_1, z_2 have the same image in $V(R)$.

Since $i : Y \rightarrow V$ and $j : Y' \rightarrow V$ are monomorphisms, the maps $Y(R) \rightarrow V(R)$ and $Y'(R) \rightarrow V(R)$ are injective. Thus, because Y and Y' are separated, if y_1, y_2 both belong to $Y(S)$, or both belong to $Y'(S)$, then $y_1 = y_2$. Otherwise we may suppose that $y_1 \in Y(S)$ and $y_2 \in Y'(S)$, so that $z_1 \in Y(R)$ and $z_2 \in Y'(R)$. Since $i \circ z_1 = j \circ z_2$, the morphism

$$z_1 \boxtimes z_2 : R \longrightarrow Y \times Y'$$

factors through $K = X$. Moreover, since X is a sheaf, the restriction map identifies $X(S)$ with $X(R)$. Hence there is an $x \in X(S)$ such that

$$u(x) \circ \tau = z_1 = y_1 \circ \tau, \quad u'(x) \circ \tau = z_2 = y_2 \circ \tau.$$

Since Y and Y' are separated, $u(x) = y_1$ and $u'(x) = y_2$, and therefore $v_1 = v_2$. This proves that V is separated.

Finally, we prove (ii). First observe that the morphism of presheaves

$$X \xrightarrow{u \boxtimes v} K,$$

where K is the kernel of the pair

$$w \circ \text{pr}_i : Y \times Y \rightrightarrows Z,$$

is an isomorphism. Indeed, for every $T \in \text{Ob } \mathcal{C}$, $X(T)$ is an equivalence relation on $Y(T)$, and hence the diagram

$$X(T) \xrightarrow{u \boxtimes v} Y(T) \times Y(T) \begin{array}{c} \xrightarrow{w \circ \text{pr}_1} \\ \xrightarrow{w \circ \text{pr}_2} \end{array} Z(T)$$

is exact in the category of sets.

Now let $S \in \text{Ob } \mathcal{C}$, and let $g_1, g_2 : S \rightarrow Z$ be two morphisms that agree on a refinement $\tau : R \hookrightarrow S$. By the construction of Z ,

$$Z(S) = Y(S)/X(S),$$

so there are morphisms $f_1, f_2 : S \rightarrow Y$ such that $wf_i = g_i$. We have $wf_1\tau = wf_2\tau$; by the preceding observation, there is a morphism $\phi : R \rightarrow X$ such that

$$u\phi = f_1\tau, \quad v\phi = f_2\tau.$$

Since X is a sheaf, there is a morphism $\psi : S \rightarrow X$ with $\phi = \psi\tau$. Thus, in $Y(R)$,

$$u\psi\tau = f_1\tau, \quad v\psi\tau = f_2\tau.$$

Because $Y(S) \rightarrow Y(R)$ is injective, Y being separated, it follows that $u\psi = f_1$ and $v\psi = f_2$, hence $g_1 = g_2$. Thus Z is separated.

Lemma 4.4.6.³⁴

³⁴Editor's note: the statement of the lemma and its proof have been given in detail.

- (i) Let $\{F_i \xrightarrow{f_i} F\}$ be a family of morphisms of presheaves, and let $G \subset F$ be its presheaf image. Then the following diagram in $\mathcal{C}$ is exact:

$$\left(\coprod_i F_i\right) \times_G \left(\coprod_j F_j\right) \xrightarrow[\text{pr}_2]{\text{pr}_1} \coprod_i F_i \longrightarrow G.$$

Equivalently, for every presheaf H , the following diagram of sets is exact:

$$\text{Hom}(G, H) \longrightarrow \prod_i \text{Hom}(F_i, H) \xrightarrow[p]{q} \prod_{i,j} \text{Hom}(F_i \times_G F_j, H). \quad (*)$$

- (ii) Every covering family of morphisms of sheaves is a universal effective epimorphic family.

Proof. For (i), let H be a presheaf. The map

$$f^* : \text{Hom}(G, H) \longrightarrow \prod_i \text{Hom}(F_i, H), \quad \phi \longmapsto (\phi \circ f_i)_i,$$

is injective. Indeed, for each $S \in \text{Ob } \mathcal{C}$, the map $\phi(S)$ is determined by the maps $(\phi \circ f_i)(S)$, since the family

$$f_i(S) : F_i(S) \longrightarrow G(S)$$

is surjective. It is clear that the image of f^* is contained in $\ker(p, q)$.

Conversely, let $\phi_i : F_i \rightarrow H$ be a family of morphisms such that, for every i, j , the diagram

$$\begin{array}{ccc} F_i \times_G F_j & \longrightarrow & F_j \\ \downarrow & & \downarrow \phi_j \\ F_i & \xrightarrow{\phi_i} & H \end{array}$$

commutes. For every S , the map

$$\prod_i F_i(S) \longrightarrow H(S)$$

then factors uniquely through a map $\phi(S) : G(S) \rightarrow H(S)$. These maps define a morphism $\phi : G \rightarrow H$ satisfying $f^*(\phi) = (\phi_i)$. This proves the exactness of $(*)$, and hence (i).

We prove (ii). Since covering families are stable under base change, it is enough to show that every covering family is effective epimorphic. Let $\{F_i \rightarrow F\}$ be a covering family of morphisms of sheaves, and let G be its presheaf image. Because the family is covering, the monomorphism $G \hookrightarrow F$ is covering as well; therefore, by 4.3.12,

$$a(G) = F.$$

Moreover, since $G \hookrightarrow F$ is a monomorphism, the fiber products $F_i \times_G F_j$ and $F_i \times_F F_j$ coincide. Part (i) therefore shows that, for every presheaf H , the diagram

$$\text{Hom}(G, H) \longrightarrow \prod_i \text{Hom}(F_i, H) \xrightarrow[p]{q} \prod_{i,j} \text{Hom}(F_i \times_F F_j, H) \quad (**)$$

is exact. If, in addition, H is a sheaf, then

$$\text{Hom}(G, H) = \text{Hom}(a(G), H) = \text{Hom}(F, H).$$

Thus (**) shows that $\{F_i \rightarrow F\}$ is an effective epimorphic family in the category $\widetilde{\mathcal{C}}$ of sheaves.

Lemma 4.4.7. *Every family of morphisms of sheaves $\{F_i \rightarrow F\}$ factors as a covering family $\{F_i \rightarrow G\}$ followed by a monomorphism $G \rightarrow F$.*

Indeed, it is enough to take G to be the sheaf image of the given family.

Proof of Proposition 4.4.3. We saw in 4.4.6 that (iii) implies (ii), while (ii) evidently implies (i). Finally, let $\{F_i \rightarrow F\}$ be an epimorphic family. By Lemma 4.4.7 it factors as a covering family followed by a monomorphism. The latter monomorphism, being dominated³⁵ by an epimorphic family, is itself an epimorphism; hence it is an isomorphism by 4.4.4.

Remark 4.4.8.³⁶ Because the presheaf image of the family $\mathcal{F} = \{F_i \rightarrow F\}$ is separated, the construction of the associated sheaf shows that the conditions of Proposition 4.4.3 are also equivalent to the following:

- (v) for every $S \in \text{Ob } \mathcal{C}$, each $f \in F(S)$ lies locally in the image of $\mathcal{F}$; that is,
- (vi) for every $S \in \text{Ob } \mathcal{C}$ and every $f \in F(S)$, the set of morphisms $S' \rightarrow S$ such that the image of f in $F(S')$ belongs to the image of one of the $F_i(S')$ is a refinement of S ;
- (vii) if the topology is defined by a pretopology $S \mapsto \mathcal{R}(S)$, then, for every $S \in \text{Ob } \mathcal{C}$ and every $f \in F(S)$, there is a family $\{S_j \rightarrow S\} \in \mathcal{R}(S)$ such that, for each j , the image f_j of f in $F(S_j)$ belongs to the image of one of the $F_i(S_j)$.

Remark 4.4.8 bis. If the sheaf F is representable, the preceding conditions are also equivalent to:

- (viii) the set of morphisms $T \rightarrow F$, with $T \in \text{Ob } \mathcal{C}$, for which there exist an i and a commutative diagram

$$\begin{array}{ccc} & F_i & \\ & \downarrow & \\ T & \longrightarrow & F \end{array}$$

is a refinement of F .

Indeed, if (viii) holds, the presheaf image of the F_i is dominated by³⁷ a refinement of F , which implies that the family is covering. Conversely, apply (vi) to $\text{id}_F \in F(F)$.

In pictorial language, this condition says: *locally on F , there is an i for which $F_i \rightarrow F$ has a section*. In particular, a morphism

$$G \longrightarrow F,$$

where G is a sheaf and F is a representable sheaf, is covering if and only if it has a section locally on F .

The following lemmas will be useful in VI_B, 8.1 and 8.2.³⁸

Lemma 4.4.8.1. *Let $Q \subset P$ be presheaves of groups on $\mathcal{C}$, with Q normal in P . Suppose that P is separated and that Q is a sheaf. Then the presheaf of groups P/Q is separated. Consequently,*

$$a(P/Q) = L(P/Q) = \varinjlim_{R \in J(S)} (P/Q)(R).$$

³⁵Editor's note: recall (cf. editor's note 17) that a morphism $g : G \rightarrow H$ is said to be *dominated* by a family of morphisms $F_i \rightarrow H$ if that family factors through g .

³⁶Editor's note: the numbering of the original, which had two paragraphs numbered 4.4.7, has been corrected to agree with later references.

³⁷Editor's note: "contains" has been replaced by "is dominated by"; cf. editor's note 17.

³⁸Editor's note: Lemmas 4.4.8.1 and 4.4.8.2 have been added.

It is enough to prove the first assertion, since the second follows from it by 4.3.11(iv) and (ii). Let $S \in \text{Ob } \mathcal{C}$, let R be a sieve of S , and let $y \in P(S)/Q(S)$ have identity image in $(P/Q)(R)$. We must show that $y = 1$.

By hypothesis, the morphism $P(S) \rightarrow P(R)$ is injective, while $Q(S) \rightarrow Q(R)$ is an isomorphism, and the upper row in the following commutative diagram is exact:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & Q(S) & \longrightarrow & P(S) & \longrightarrow & P(S)/Q(S) & \longrightarrow & 1 \\ & & \downarrow \sim & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & Q(R) & \longrightarrow & P(R) & \longrightarrow & (P/Q)(R) & & \end{array}$$

The result follows if the lower row is exact. Let $f : R \rightarrow P$ have identity image in $(P/Q)(R)$, and let $\phi : T \rightarrow R$ with $T \in \text{Ob } \mathcal{C}$. Then $f \circ \phi$ is the identity in

$$(P/Q)(T) = P(T)/Q(T),$$

that is, $f \circ \phi \in Q(T)$. Hence $\phi \mapsto f \circ \phi$ is a functorial map $R(T) \rightarrow Q(T)$, and so defines a morphism of functors $\pi : R \rightarrow Q$ such that $\pi(\text{id}_R) = f$. Thus $f \in Q(R)$. This proves the exactness of the lower row and establishes the lemma.

Lemma 4.4.8.2. *Let $H \subset G$ be presheaves of groups on $\mathcal{C}$.*

- (i) *If H is normal in G , then $L(H)$ is normal in $L(G)$.*
- (ii) *If H is central in G , then $L(H)$ is central in $L(G)$.*

Let $S \in \text{Ob } \mathcal{C}$. We must show (cf. I 2.3.6) that $L(H)(S)$ is normal, respectively central, in $L(G)(S)$. Let

$$h \in L(H)(S), \quad g \in L(G)(S).$$

There are a sieve R of S and elements $h' \in H(R)$, $g' \in G(R)$ such that

$$h = z_R(h'), \quad g = z_R(g')$$

in the notation of 4.3.10. Since z_R is a group morphism,

$$ghg^{-1}h^{-1} = z_R(g'h'g'^{-1}h'^{-1}).$$

In case (i),

$$g'h'g'^{-1}h'^{-1} \in H(R),$$

and therefore $ghg^{-1}h^{-1} \in L(H)(S)$. In case (ii), $g'h'g'^{-1}h'^{-1} = 1$, and hence $ghg^{-1}h^{-1} = 1$.

Proposition 4.4.9. *Every equivalence relation in $\tilde{\mathcal{C}}$ is universally effective (3.3.3). More precisely, let R be a $\tilde{\mathcal{C}}$ -equivalence relation on the sheaf X . Then the sheaf associated with the separated presheaf*

$$i(X)/i(R) : S \longmapsto X(S)/R(S)$$

is a universal effective quotient of X by R .

Let X/R denote the sheaf quotient of X by R , which exists by 4.4.1(ii):

$$X/R = a(i(X)/i(R)).$$

We must show that $X \rightarrow X/R$ is a universal effective epimorphism and that the morphism

$$f : R \longrightarrow X \times_{X/R} X$$

is an isomorphism. The first assertion has already been proved in 4.4.3. As for f , it is obtained by applying the functor a to the morphism

$$i(R) \longrightarrow i(X) \times_{i(X/R)} i(X),$$

or, since $i(X)/i(R)$ is separated by 4.4.5(ii), so that

$$i(X)/i(R) \longrightarrow i(X/R)$$

is a monomorphism, to the canonical morphism

$$i(R) \longrightarrow i(X) \times_{i(X)/i(R)} i(X).$$

Thus we are reduced to proving the same statement in the category of presheaves. But $i(X)/i(R)$ is the presheaf

$$S \longmapsto X(S)/R(S),$$

so the assertion reduces to its set-theoretic analogue, where it is immediate.

Proposition 4.4.10. *Under the hypotheses of 4.4.9, let Y be a subsheaf of X , and write R_Y for the equivalence relation induced on Y by R . Then the canonical morphism (3.1.6)*

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. It identifies Y/R_Y with a subsheaf of X/R , namely the sheaf image of the composite

$$Y \longrightarrow X \longrightarrow X/R.$$

The morphism of presheaves

$$i(Y)/i(R_Y) = i(Y)/i(R)_{i(Y)} \longrightarrow i(X)/i(R)$$

is a monomorphism. Since the functor a is left exact (4.3.16), it carries monomorphisms to monomorphisms; hence

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. The final assertion follows from the commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y/R_Y & \longrightarrow & X/R \end{array}$$

and the fact that $Y \rightarrow Y/R_Y$ is covering.

In view of this proposition, we shall always identify Y/R_Y with a subsheaf of X/R .

Proposition 4.4.11. *Let R be a $\mathcal{C}$ -equivalence relation on the sheaf X . For every subsheaf Y of X stable under R , let $Y' = Y/R_Y$, regarded as a subsheaf of $X' = X/R$. Then*

$$Y = Y' \times_{X'} X,$$

and the maps

$$Y \longmapsto Y/R_Y, \quad Y' \longmapsto Y' \times_{X'} X$$

define a bijective correspondence between the subsheaves Y of X stable under R and the subsheaves Y' of X' .

If Y' is a subsheaf of X' , then $Y' \times_{X'} X$ is a subsheaf of X stable under R .³⁹ If Y' is obtained by passage to the quotient from a subsheaf Y of X , then Y is a subobject of $Y' \times_{X'} X$. It is therefore enough to prove the following: if Y and Y_1 are two subsheaves of X , both stable under R , with Y_1 containing Y , and if their quotients Y/R_Y and Y_1/R_{Y_1} are identical, then $Y = Y_1$.

We may plainly reduce to the case $Y_1 = X$. Write

$$P = i(X)/i(R), \quad Q = i(Y)/i(R_Y)$$

for the corresponding presheaves. The diagram

$$\begin{array}{ccc} Y & \longrightarrow & Q \\ \downarrow & & \downarrow \\ X & \longrightarrow & P \end{array}$$

is Cartesian. On the other hand, there is a commutative diagram

$$\begin{array}{ccc} Q & \longrightarrow & a(Q) \\ \downarrow & & \downarrow \sim \\ P & \longrightarrow & a(P). \end{array}$$

Since $Q \hookrightarrow a(Q)$ is covering by 4.3.11, the monomorphism $Q \hookrightarrow P$ is covering, so Q is a refinement of P . After base change, Y is a refinement of X . Because X and Y are sheaves, 4.3.12 then gives $Y = X$.

4.4.12. In particular, if Y is a subsheaf of X and $Y' = Y/R_Y$, the preceding correspondence defines a subsheaf $\bar{Y}$ of X , stable under R , containing Y , and minimal with these properties. It is called the *saturation* of Y for the equivalence relation R .

4.5. The Case of a Topology Coarser than the Canonical Topology

By 4.3.6 and 4.3.8, the following conditions on a topology $\mathcal{T}$ on $\mathcal{C}$ are equivalent:

- (i) $\mathcal{T}$ is coarser than the canonical topology of $\mathcal{C}$;
- (ii) every representable presheaf is a sheaf for $\mathcal{T}$;
- (iii) every covering sieve for $\mathcal{T}$ is universal effective epimorphic.

If $\mathcal{T}$ is defined by a pretopology $S \mapsto \mathcal{R}(S)$, these conditions are also equivalent to:

- (iv) every family belonging to $\mathcal{R}(S)$ is universal effective epimorphic.

When these conditions hold, the canonical functor $\mathcal{C} \rightarrow \hat{\mathcal{C}}$ factors through a functor

$$j_{\mathcal{C}} = j : \mathcal{C} \longrightarrow \tilde{\mathcal{C}};$$

we also write $j(S) = \tilde{S}$.⁴⁰

Proposition 4.5.1. *The functor $j : \mathcal{C} \rightarrow \tilde{\mathcal{C}}$ is fully faithful and commutes with arbitrary inverse limits. In particular, it is left exact and therefore preserves algebraic structures defined by finite inverse limits.*

³⁹Editor's note: moreover, $(Y' \times_{X'} X)/R = Y'$.

⁴⁰Editor's note: we shall also write $\mathbf{h}_S = \tilde{S}$; cf. the first commutative diagram in 4.5.4.

This follows immediately from the commutative factorization

$$\begin{array}{ccc} & \mathcal{C} & \\ j \swarrow & & \searrow \text{canonical} \\ \mathcal{C} & \xrightarrow{i} & \widehat{\mathcal{C}} \end{array}$$

and from 4.4.1(i).

Before giving further properties of j , we must define the topology induced on a category $\mathcal{C}/S$. We no longer suppose here that the given topology is necessarily coarser than the canonical topology. A sieve of an object T in $\mathcal{C}$, together with a morphism $T \rightarrow S$, naturally defines a sieve of T in $\mathcal{C}/S$, because the definition of a sieve of T depends only on the category

$$\mathcal{C}/T = (\mathcal{C}/S)/T.$$

For example, if the sieve is defined by a family $\{T_i \rightarrow T\}$, its image in $\mathcal{C}/S$ is defined by the same family, now regarded as a family of morphisms in $\mathcal{C}/S$.

Declaring these images of covering sieves to be covering defines a topology on $\mathcal{C}/S$, called the *topology induced* by the given topology. In the terminology of [AS], 2.3, it is the coarsest topology on $\mathcal{C}/S$ for which the canonical functor

$$i_S : \mathcal{C}/S \longrightarrow \mathcal{C}$$

is a comorphism.⁴¹ By definition, the identifications

$$(\mathcal{C}/S)/T = \mathcal{C}/T$$

respect these topologies.

Proposition 4.5.2. *Let S be a representable sheaf on $\mathcal{C}$, and let $F \rightarrow S$ be a morphism in $\widehat{\mathcal{C}}$. The presheaf*

$$S' \longmapsto \text{Hom}_S(S', F)$$

on $\mathcal{C}/S$ is separated (respectively, is a sheaf) if and only if F is a separated presheaf (respectively, a sheaf) on $\mathcal{C}$.

For every functor P , one has (I, 1.4.1)

$$\text{Hom}(P, F) = \coprod_{f \in \text{Hom}(P, S)} \text{Hom}_f(P, F).$$

⁴² Let $S' \in \text{Ob}(\mathcal{C})$, and let $\eta : C' \rightarrow S'$ be a covering sieve. Since S is a sheaf, the map $f \mapsto f \circ \eta$ is a bijection

$$\text{Hom}(S', S) \xrightarrow{\sim} \text{Hom}(C', S).$$

Consequently, the map

$$\text{Hom}_{\mathcal{C}}(S', F) \longrightarrow \text{Hom}_{\widehat{\mathcal{C}}}(C', F)$$

decomposes as the “disjoint union” of the maps

$$\text{Hom}_f(S', F) \longrightarrow \text{Hom}_{f \circ \eta}(C', F).$$

The proposition follows.

⁴¹Editor’s note: that is, for every object $T \rightarrow S$ of $\mathcal{C}/S$, every covering sieve R' of $i_S(T \rightarrow S) = T$, when regarded as a sieve of the object $(T \rightarrow S) \in \text{Ob}(\mathcal{C}/S)$, is covering.

⁴²Editor’s note: the argument implicit in the original has been expanded below.

Corollary 4.5.3. *If a topology on $\mathcal{C}$ is coarser than the canonical topology, then the topology it induces on $\mathcal{C}/S$ is coarser than the canonical topology of $\mathcal{C}/S$.*

Corollary 4.5.4. *Suppose that the given topology on $\mathcal{C}$ is coarser than the canonical topology. For every $S \in \text{Ob}(\mathcal{C})$, there is an equivalence of categories*

$$\widetilde{\mathcal{C}}_{/\widetilde{S}} \xrightarrow{\sim} \widehat{\mathcal{C}}/S.$$

The following diagrams commute up to isomorphism; every unlabelled arrow in them is an equivalence:

$$\begin{array}{ccccccc} \mathcal{C}/S & \xrightarrow{(j_{\mathcal{C}})_{/S}} & \widetilde{\mathcal{C}}_{/\widetilde{S}} & \xrightarrow{(i_{\mathcal{C}})_{/\widetilde{S}}} & \widehat{\mathcal{C}}_{/\widehat{S}} & \xrightarrow{(a_{\mathcal{C}})_{/\widehat{S}}} & \widetilde{\mathcal{C}}_{/\widetilde{S}} \\ \parallel & & \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ \mathcal{C}/S & \xrightarrow{j_{\mathcal{C}/S}} & \widehat{\mathcal{C}}/S & \xrightarrow{i_{\mathcal{C}/S}} & \widehat{\mathcal{C}}/S & \xrightarrow{a_{\mathcal{C}/S}} & \widehat{\mathcal{C}}/S. \end{array}$$

If $T \rightarrow S$ is an object of $\mathcal{C}/S$, one also has the commutative diagram

$$\begin{array}{ccccc} (\widetilde{\mathcal{C}}_{/\widetilde{S}})_{/\widetilde{T}} & \longrightarrow & (\widehat{\mathcal{C}}/S)_{/\widetilde{T}} & \searrow & (\widehat{\mathcal{C}}/S)_{/T} \\ \downarrow \wr & & & & \uparrow \\ \widetilde{\mathcal{C}}_{/\widetilde{T}} & \longrightarrow & \widehat{\mathcal{C}}/T & \xrightarrow{\sim} & (\widehat{\mathcal{C}}/S)_{/T}. \end{array}$$

The commutativity of the first two squares follows from the definition of the equivalence $\widetilde{\mathcal{C}}_{/\widetilde{S}} \rightarrow \widehat{\mathcal{C}}/S$. To prove the commutativity of the last square, it is enough to observe that the following square commutes:

$$\begin{array}{ccc} \widehat{\mathcal{C}}_{/\widehat{S}} & \xrightarrow{L'} & \widehat{\mathcal{C}}_{/\widehat{S}} \\ \downarrow & & \downarrow \\ \widehat{\mathcal{C}}/S & \xrightarrow{L''} & \widehat{\mathcal{C}}/S. \end{array}$$

Here L' is the restriction of the functor $L_{\mathcal{C}}$ to $\widehat{\mathcal{C}}_{/\widehat{S}}$, and L'' is the functor $L_{\mathcal{C}/S}$. This is seen immediately by returning to the definition of the functors L , given after 4.3.9. The second diagram is simply the restriction to the categories of sheaves of the corresponding diagram for the categories of presheaves (Exposé I, no. 1), which is commutative.

Scholium 4.5.5. The preceding assertions show that, when the given topology is coarser than the canonical topology, one may identify $\mathcal{C}$ with a full subcategory of $\widehat{\mathcal{C}}$, itself a full subcategory of $\widehat{\mathcal{C}}$. Under these identifications one may make the customary abuses of language concerning $\mathcal{C} \rightarrow \widehat{\mathcal{C}}$, justified by the preceding commutative diagrams. In particular, the first diagram of 4.5.4 shows that the functor a may be used without any special precaution.

In the next subsection we shall see that the identification of $\mathcal{C}$ with a full subcategory of $\widehat{\mathcal{C}}$, unlike the corresponding identification with $\widehat{\mathcal{C}}$, commutes with the formation of certain direct limits; we shall then explain how this fact is to be used.

From now on, unless expressly stated otherwise, we assume that the given topology is coarser than the canonical topology, and we systematically make the identifications described above.

Proposition 4.5.6. *Let F and G be two sheaves over S , and let $f : F \rightarrow G$ be an S -morphism. The following conditions on f are equivalent:*

- (i) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in $\mathcal{C}$;*
- (ii) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in $\mathcal{C}/S = \widehat{\mathcal{C}}/S$.*

For monomorphisms and isomorphisms this is immediate, since it is a question about presheaves. For epimorphisms it follows from the description of epimorphisms as covering morphisms (4.4.3) and from the fact that, by the definition of the induced topology, these are the same in $\widetilde{\mathcal{C}}$ and in $\widetilde{\mathcal{C}}/S$.

Proposition 4.5.7. *Let $f : F \rightarrow G$ be a morphism of sheaves. The following conditions are equivalent:*

- (i) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism);*
- (ii) *for every $S \in \text{Ob}(\mathcal{C})$, the morphism $f_S : F_S \rightarrow G_S$ is locally a monomorphism (respectively, an epimorphism, respectively, an isomorphism), that is:*
- (iii) *for every $S \in \text{Ob}(\mathcal{C})$, the set of morphisms $T \rightarrow S$ for which $F_T \rightarrow G_T$ is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) is a refinement of S .*

If the given topology is defined by a pretopology $\mathcal{R}$, these conditions are also equivalent to:

- (iv) *for every $S \in \text{Ob}(\mathcal{C})$, there is a covering family $\{S_i \rightarrow S\} \in \mathcal{R}(S)$ such that, for every i , $F_{S_i} \rightarrow G_{S_i}$ is a monomorphism (respectively, an epimorphism, respectively, an isomorphism).*

If $\mathcal{C}$ has a final object e , it is enough in conditions (ii), (iii), and (iv) to take $S = e$.

Clearly (ii), (iii), and (iv) are equivalent. To prove the equivalence of (i) and (ii), as well as the assertion concerning a final object, it is enough to show that (ii) implies (i) and that the properties in question are stable under base change. We prove the latter point first. For monomorphisms and isomorphisms it is immediate, since it is a question about presheaves. For epimorphisms it follows from the fact that every epimorphism of sheaves is universal (4.4.3).

It remains to show that (ii) implies (i). Suppose first that $f_S : F_S \rightarrow G_S$ is locally a monomorphism (respectively, an isomorphism). There is then a covering sieve C of S such that f_T is a monomorphism (respectively, an isomorphism) for every $T \rightarrow S$ belonging to C . Since an inverse limit of monomorphisms (respectively, isomorphisms) is again one, $\text{Hom}(C, f)$ is a monomorphism (respectively, an isomorphism); see 4.1.4. The commutative diagram

$$\begin{array}{ccc} \text{Hom}(C, F) & \xrightarrow{\text{Hom}(C, f)} & \text{Hom}(C, G) \\ \uparrow \wr & & \uparrow \wr \\ F(S) & \xrightarrow{f(S)} & G(S) \end{array}$$

therefore shows that $f(S)$ is injective (respectively, bijective).

Finally, suppose that $f : F \rightarrow G$ is locally an epimorphism, and let $H \subset G$ be its image. For every $S \rightarrow G$, the sheaf $H \times_G S$ is the image of

$$f \times_G S : F \times_G S \longrightarrow S.$$

To prove that f is an epimorphism, it is enough to prove that H is a refinement of G , or equivalently that $H \times_G S$ is a refinement of S for every S . But this holds after each base change $T \rightarrow S$ belonging to a refinement of S , since $f \times_G S$ is locally covering. Hence $H \times_G S$ is itself a refinement of S , by Axiom (T 2).

Corollary 4.5.8. *Let F and G be two sheaves over S , and let $f : F \rightarrow G$ be an S -morphism. Then f is a monomorphism, respectively an epimorphism, respectively an isomorphism, if and only if it is so locally on S .*

Remark 4.5.9. The proof of the proposition shows that its assertion about monomorphisms (respectively, isomorphisms) remains valid if one assumes only that F is a separated presheaf (respectively, a sheaf) and G is an arbitrary presheaf (respectively, a separated presheaf).

We return temporarily to the case of an arbitrary topology and make the following definition.

Definition 4.5.10. Let $G \rightarrow F$ be a morphism in $\widehat{\mathcal{C}}$. We say that G is a relative sheaf over F if, for every F -functor H and every refinement R of H , the canonical map

$$\mathrm{Hom}_F(H, G) \xrightarrow{\sim} \mathrm{Hom}_F(R, G) \quad (+)$$

is bijective.

Proposition 4.5.2 immediately gives the following generalization.

Proposition 4.5.11. If F is a sheaf, then G is a relative sheaf over F if and only if G is a sheaf.

Lemma 4.5.12. In a sequence

$$X \longrightarrow T \longrightarrow S,$$

where X, T, S are three objects of $\widehat{\mathcal{C}}$, suppose that X is a relative sheaf over T . Then

$$U = \prod_{T/S} X$$

is a relative sheaf over S .

Indeed, for every S -functor Y , one has

$$\mathrm{Hom}_S(Y, U) = \mathrm{Hom}_T(T \times_S Y, X).$$

If C is a sieve of Y , then $T \times_S C$ is a sieve of $T \times_S Y$, and the conclusion is immediate.

Corollary 4.5.13. The presheaves $\mathrm{Hom}_{T/S}(X, Y)$, $\mathrm{Isom}_S(X, Y)$, and the analogous presheaves are sheaves whenever all the arguments occurring in them are sheaves.

Indeed, all these presheaves are constructed using fiber products and presheaves $\prod_{T/S} X$ (I, 1.7 and II, 1). It is therefore enough to verify the assertion for a presheaf $\prod_{T/S} X$; in that case it follows from 4.5.11 and 4.5.12.

4.6. Description of the Quotient of a Sheaf by an Equivalence Relation

Recall that the given topology $\mathcal{T}$ is assumed to be coarser than the canonical topology.

Proposition 4.6.1. Let $R \xrightarrow[p_2]{p_1} X$ be a $\widetilde{\mathcal{C}}$ -equivalence relation in the sheaf X . Define $F \in \mathrm{Ob}(\widehat{\mathcal{C}})$ as follows: for each $S \in \mathrm{Ob}(\mathcal{C})$,⁴³

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ whose quotient} \\ \text{by } R_Z \text{ is } S, \text{ that is, for which } R_Z \rightrightarrows Z \rightarrow S \text{ is exact} \end{array} \right\}.$$

For every sheaf Y , $\mathrm{Hom}(Y, F)$ is naturally identified with the set

$$\left\{ \begin{array}{l} \text{sub-}Y\text{-sheaves of } X \times Y, \text{ stable under } R \times Y, \\ \text{and having quotient } Y \end{array} \right\}.$$

⁴³Editor's note: $R \times S$ denotes the equivalence relation in $X \times S$ defined by $R \times S_{\mathrm{diag}} \subset X \times X \times S \times S$, and R_Z denotes the equivalence relation that it induces on Z (cf. 3.1.6).

In particular, the subsheaf $R \subset X \times X$ corresponds to an element $p \in \text{Hom}(X, F)$, and the exact diagram

$$R \begin{array}{c} \xrightarrow{p_1} \\ \xrightarrow{p_2} \end{array} X \xrightarrow{p} F$$

identifies F with the sheaf quotient X/R .

Indeed, put $Q = X/R$. For every sheaf Y and every morphism $f \in \text{Hom}(Y, Q)$, corresponding to a section $s : Y \hookrightarrow Q \times Y$, consider the diagram

$$\begin{array}{ccccc} R \times Y & \rightrightarrows & X \times Y & \longrightarrow & Q \times Y \\ & & \uparrow & & \uparrow_s \\ & & Z & \longrightarrow & Y, \end{array} \quad (*)$$

in which the right-hand square is cartesian. By 4.4.11, Z is immediately seen to be a sub- Y -sheaf of $X \times Y$, stable under $R \times Y$, with quotient Y . Conversely, every Z of this kind arises from a unique section of $Q \times Y$ over Y . Taking Y first to be representable gives an isomorphism $Q \simeq F$; taking arbitrary Y then gives the asserted description of $\text{Hom}(Y, F)$. Finally, the canonical morphism $X \rightarrow Q$ corresponds to the sub- X -sheaf $R \subset X \times X$, which completes the proof.

Corollary 4.6.2. *Let G be any subfunctor of F such that $\text{Hom}(X, G) \subset \text{Hom}(X, F)$ contains R . Then the canonical morphism $p : X \rightarrow F$ factors through G .*

Since p is covering (4.4.9 and 4.4.3), it follows that G is a refinement of F . In particular, every subsheaf G of F satisfying the preceding condition equals F (4.3.12).

4.6.3. We now consider the case in which X and R are representable. We begin with some terminology. Besides the conditions (a)–(d) introduced in 3.4.1, we shall use the following conditions on a family (M) of morphisms of $\mathcal{C}$; for completeness, we repeat conditions (a)–(c):

(a) (M) is stable under base change.

(b) The composite of two elements of (M) belongs to (M) .

(c) Every isomorphism belongs to (M) .

($d_{\mathcal{T}}$) Every element of (M) is covering. ⁴⁴

($e_{\mathcal{T}}$) Let $f : X \rightarrow Y$ be a morphism of $\mathcal{C}$. If there is a refinement R of Y such that, for every $Y' \rightarrow R$, the morphism $X \times_Y Y' \rightarrow Y'$ belongs to (M) , then f belongs to (M) .

Conditions (a) and (b) imply

(a') the fiber product of two elements of (M) belongs to (M) .

On the other hand, by 4.3.9, conditions (a) and ($d_{\mathcal{T}}$) imply

(d') every element of (M) is a universal effective epimorphism.

4.6.4. When $\mathcal{C}$ has fiber products, the family of covering morphisms, denoted $(M_{\mathcal{T}})$, satisfies the preceding conditions. Indeed (cf. 4.2.3), (a) follows from (C 1), (b) from (C 2), (c) from (C 4), ($d_{\mathcal{T}}$) from the definition, and ($e_{\mathcal{T}}$) from (C 5). Thus the results established below for a family satisfying these conditions apply in particular to $(M_{\mathcal{T}})$. In particular, $\mathcal{T}$ may be taken to be the canonical topology and (M) to be the family of universal effective epimorphisms.

⁴⁴Editor's note: $\mathcal{T}$ denotes the given topology on $\mathcal{C}$, which is coarser than the canonical topology.

Lemma 4.6.5. *Let (M) be a family of morphisms satisfying conditions (a)–(e $\mathcal{T}$) above. Let R be a $\mathcal{C}$ -equivalence relation of type (M) in $X \in \text{Ob}(\mathcal{C})$. Let $\tilde{X}$ be the sheaf defined by X , let $\tilde{R}$ be the $\mathcal{C}$ -equivalence relation in $\tilde{X}$ defined by R , and let $\tilde{X}/\tilde{R}$ be the sheaf quotient. Then R is (M) -effective if and only if $\tilde{X}/\tilde{R}$ is representable. When these conditions hold, $\tilde{X}/\tilde{R}$ is represented by the quotient X/R .*

Suppose first that R is (M) -effective, and put $Y = X/R$. The canonical morphism $p : X \rightarrow Y$ belongs to (M) , hence is covering by (d $\mathcal{T}$). The corresponding morphism

$$\tilde{p} : \tilde{X} \longrightarrow \tilde{X}/\tilde{R}$$

is therefore a universal effective epimorphism in $\tilde{\mathcal{C}}$ (4.4.3). Consequently it identifies $\tilde{X}/\tilde{R}$ with the quotient of $\tilde{X}$ by the equivalence relation R' defined by $\tilde{p}$. Since the canonical functor $\mathcal{C} \rightarrow \tilde{\mathcal{C}}$ commutes with fiber products, R' is precisely $\tilde{R}$, because R is the equivalence relation defined by p .

Conversely, suppose that $\tilde{X}/\tilde{R}$ is represented by an object Y of $\mathcal{C}$. Let $p : X \rightarrow Y$ be the morphism deduced from the canonical morphism $\tilde{X} \rightarrow \tilde{X}/\tilde{R}$; by 4.4.3 it is covering. As before, R is plainly the equivalence relation defined by p . It remains only to show that $p \in (M)$. The cartesian diagram

$$\begin{array}{ccccc} R & \xrightarrow{\sim} & X \times_Y X & \longrightarrow & X \\ p_2 \searrow & & \downarrow \text{pr}_2 & & \downarrow p \\ & & X & \xrightarrow{p} & Y \end{array}$$

shows that, after the covering base change p , the morphism p becomes p_2 , which belongs to (M) . The assertion follows from (e $\mathcal{T}$).

Corollary 4.6.5.1. ⁴⁵ *Let (M) satisfy conditions (a)–(e $\mathcal{T}$) above, and let $f : G \rightarrow G'$ be a morphism of $\mathcal{C}$ -groups such that $f \in (M)$. Suppose that $\text{Ker}(f)$ is representable (as is the case when $\mathcal{C}$ has a final object e). Then the equivalence relation in G defined by $H = \text{Ker}(f)$ is (M) -effective, and G' represents the sheaf quotient $\tilde{G}/\tilde{H}$ for the topology $\mathcal{T}$.*

This follows from 4.6.5 and 3.4.7.1.

We can now state the main result of this subsection.

Theorem 4.6.6. *Let (M) be a family of morphisms satisfying Axioms (a)–(e $\mathcal{T}$) of 4.6.3. Let R be a $\mathcal{C}$ -equivalence relation of type (M) (cf. 3.4.3) in an object X of $\mathcal{C}$. Consider the functor $F \in \text{Ob}(\hat{\mathcal{C}})$ defined by*

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S, \text{ stable under } R \times S, \\ \text{whose quotient by } R_Z \text{ is } S \end{array} \right\}.$$

Let F_0 be the subfunctor of F defined as follows: $F_0(S)$ consists of the representable $Z \in F(S)$, that is,

$$F_0(S) = \left\{ \begin{array}{l} \text{sub-}\mathcal{C}/S\text{-objects } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ such that} \\ R_Z \text{ is } (M)\text{-effective with quotient } S, \text{ i.e. such that} \\ Z \rightarrow S \text{ belongs to } (M) \text{ and } R_Z \simeq Z \times_S Z \end{array} \right\}.$$

Then:

- (i) *The morphism $p \in \text{Hom}(X, F) = F(X)$ defined by the subobject $R \subset X \times X$ identifies F with the sheaf quotient of X by R .*
- (ii) *The following conditions are equivalent:*

⁴⁵Editor's note: This corollary has been added.

- (a) F is representable;
- (b) F_0 is representable;
- (c) R is (M) -effective.

Under these conditions, $F = F_0 = X/R$.

- (iii) Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S_i \rightarrow S\}$ and every family $\{T_i \rightarrow S_i\}$ of morphisms in (N) , every descent datum on the T_i relative to $\{S_i \rightarrow S\}$ is effective. Suppose that X is squarable (cf. 1.6.0) and that $R \rightarrow X \times X$ belongs to (N) . Then $F_0 = F$.

Proof. Assertion (i) was proved in 4.6.1.

For (ii), we have already proved the equivalence of (a) and (c), as well as the equality $F = X/R$. It remains to prove that either (b) or (c) implies $F_0 = F$. First note, as stated in the theorem, that F_0 is indeed a subfunctor of F : if $S \in \text{Ob}(\mathcal{C})$ and $Z \in F_0(S)$, then $Z \rightarrow S$ is squarable, and hence $Z \times_S S' \in F_0(S')$ for every $S' \rightarrow S$. Since $R \in F(X)$ belongs to $F_0(X)$, Corollary 4.6.2 shows that (b) implies $F_0 = F$.

Now suppose that (c) holds, and let Q be an object of $\mathcal{C}$ representing X/R . Then $X \rightarrow Q$ belongs to (M) . For every $S \in \text{Ob}(\mathcal{C})$ and every $Z \in F(S)$, diagram (*) of 4.6.1 gives

$$Z = S \times_{Q \times S} (X \times S).$$

Thus Z is representable and $Z \rightarrow S$ belongs to (M) , so $Z \in F_0(S)$.

For (iii), let $f \in \text{Hom}(S, F)$ correspond to $Z \in F(S)$. We must show that f factors through F_0 , or equivalently that Z is representable. This is immediate when f factors through X , by the following lemma.

Lemma 4.6.7. *Let $x_0 \in X(S)$. The image of x_0 in $F(S)$ corresponds to the subsheaf $Z \subset X_S$ defined by the two cartesian squares*

$$\begin{array}{ccccc} X_S & \xrightarrow{\text{id}_{X_S} \times x_0} & X_S \times_S X_S & \longrightarrow & X \times X \\ \uparrow & & \uparrow & & \uparrow \\ Z & \longrightarrow & R_S & \longrightarrow & R. \end{array}$$

This follows at once from the description of the morphism $X \rightarrow F$.

Returning to the proof of the theorem, suppose first that f factors through X . Then Z is representable; moreover, since $R \rightarrow X \times X$ belongs to (N) , so does $Z \rightarrow X_S$.

In general, f need not factor through X . Since $X \rightarrow F$ is covering (4.4.3), however, 4.4.8(vii) gives a covering family $\{S_i \rightarrow S\}$ and, for each i , a morphism $S_i \rightarrow X$ making the diagram

$$\begin{array}{ccc} X & \longrightarrow & F \\ \uparrow & & \uparrow f \\ S_i & \longrightarrow & S \end{array}$$

commute. By the preceding argument, the morphism $f_i : S_i \rightarrow F$ defined by this diagram belongs to $\text{Hom}(S_i, F_0)$ and corresponds to the subsheaf $Z \times_S S_i \subset X_{S_i}$. The morphism $Z \times_S S_i \rightarrow X_{S_i}$ belongs to (N) , and the family $\{X_{S_i} \rightarrow X_S\}$ is covering. It therefore remains only to prove the following proposition.

Proposition 4.6.8. *Let $\{S_i \rightarrow S\}$ be a covering family and let Z be a sheaf over S . Suppose that, for every i , the S_i -functor $Z \times_S S_i$ is represented by an object T_i . Then the family (T_i) carries a canonical descent datum relative to $\{S_i \rightarrow S\}$. The sheaf Z is representable if and only if this datum is effective; in that case the descended object represents Z .*

First observe that, by 4.4.3, $\{S_i \rightarrow S\}$ is a universal effective epimorphic family in $\widetilde{\mathcal{C}}$, and hence a descent family in $\mathcal{C}$ (2.3). If Z is represented by an object T , then $T \times_S S_i$, regarded as a sheaf, is isomorphic to $Z \times_S S_i$. Thus the descent datum on the T_i is effective, and its uniquely determined descended object is isomorphic to T , hence represents Z .

Conversely, suppose that the canonical descent datum on the T_i is effective, and let T be the descended object. Because $\{S_i \rightarrow S\}$ is a descent family in $\widetilde{\mathcal{C}}$, there is an S -morphism $T \rightarrow Z$ whose base change to each S_i is the canonical morphism $T_i \rightarrow Z \times_S S_i$. This morphism is locally an isomorphism; since both T and Z are sheaves, 4.5.8 implies that it is an isomorphism.

Corollary 4.6.9. *Let R be an (M) -effective equivalence relation in X . For every sheaf F , the map*

$$\mathrm{Hom}(X/R, F) \longrightarrow \mathrm{Hom}(X, F)$$

identifies the first set with the subset of the second consisting of morphisms compatible with R .

Corollary 4.6.10. *Let $\mathcal{T}'$ be a topology coarser than $\mathcal{T}$, for which the morphisms in (M) are covering. Under the hypotheses of 4.6.6(iii), X/R is also the sheaf quotient of X by R for every topology lying between $\mathcal{T}'$ and the canonical topology.*

Remark 4.6.11. If, in the statement of 4.6.6(iii), one assumes in addition that the descended morphism $T \rightarrow S$ belongs to (N) whenever the hypotheses there are satisfied, then the inclusion morphisms $Z \hookrightarrow X_S$ also belong to (N) . This follows immediately from the construction of Z by descent.

Remark 4.6.12. The implications

$$(c) \implies (b) \implies (a) \quad \text{and} \quad (c) \implies [F_0 = F = X/R]$$

were proved without using the sufficiency direction of Lemma 4.6.5, the only place where condition $(e_{\mathcal{T}})$ enters. They therefore remain valid when (M) satisfies only conditions (a)– $(d_{\mathcal{T}})$. The family of squarable covering morphisms is an example of such a family (M) (compare 4.6.4). For the canonical topology these are precisely the universal effective epimorphisms. Hence:

Corollary 4.6.13. *Let R be a universal effective equivalence relation in X . Then the object X/R of $\mathcal{C}$ is the sheaf quotient of X by R for the canonical topology. It represents the following functor: $(X/R)(S)$ is the set of sub- $\mathcal{C}/S$ -objects Z of X_S , stable under $R \times S$, for which the induced equivalence relation is universal effective and has quotient S .*

Similarly, for an arbitrary topology:

Corollary 4.6.14. *Let (M) be the family of squarable covering morphisms. If R is an (M) -effective equivalence relation in X , then the object X/R of $\mathcal{C}$ is the sheaf quotient of X by R , and it represents the functor F_0 of 4.6.6.*

Scholium 4.6.15. We can now make the following clarifications to 4.5.5. In questions involving inverse limits alone—fiber products, algebraic structures, and so forth—the results of Exposé I and 4.5.5 allow us to identify $\mathcal{C}$, without distinction, with a full subcategory of either $\mathcal{C}$ or $\widetilde{\mathcal{C}}$. The same is not true for questions that mix inverse and direct limits.

Whenever both inverse and direct limits occur, and especially when passing to quotients (for example, when putting a group structure on the quotient of a group by an invariant subgroup), we shall regard the given category as embedded in the category of sheaves. Thus, if R is a $\mathcal{C}$ -equivalence relation in an object X of $\mathcal{C}$, the notation X/R will mean the sheaf quotient of X by R , formerly denoted $j(X)/j(R)$, and, when this sheaf is representable, also

the object that represents it. The preceding results show that, in the most important cases, a quotient in $\mathcal{C}$ is also a quotient in the category of sheaves. In any event, we shall not use the notation X/R for a quotient in $\mathcal{C}$ that does not coincide with the quotient in $\widetilde{\mathcal{C}}$ —for instance, one that is not universal. This modifies the definitions of no. 3.

To study a problem of this kind, one therefore works first in the category of sheaves, where all the usual results hold (cf. no. 4.4), and then specializes the resulting statements to the original category, using the results of the present subsection and, when available, criteria for the effectivity of descent. Examples of this method will appear in the following subsections.

4.7. Use of Effectivity Criteria: Isomorphism Theorem

In this subsection we give an example of the use of effectivity criteria. The initial data are a topology $\mathcal{T}$ on $\mathcal{C}$, still assumed coarser than the canonical topology; a family (M) of morphisms of $\mathcal{C}$ satisfying Axioms (a)– $(e_{\mathcal{T}})$ of 4.6.3; and a family (N) of morphisms of $\mathcal{C}$ which may satisfy the following axioms:

- (a) (N) is stable under base change;
- $(f_{\mathcal{T}})$ morphisms in (N) descend for the given topology. More precisely, for every $S \in \text{Ob}(\mathcal{C})$, every covering family $\{S_i \rightarrow S\}$, and every family $\{T_i \rightarrow S_i\}$ of morphisms in (N) , every descent datum on the T_i relative to $\{S_i \rightarrow S\}$ is effective; if T denotes the descended object, then $T \rightarrow S$ belongs to (N) .

Since every element of (M) is covering (condition 4.6.3 $(d_{\mathcal{T}})$), Axiom $(f_{\mathcal{T}})$ implies the following axiom:⁴⁶

- (f_M) if $Y \rightarrow X$ belongs to (N) and $X \rightarrow X_1$ belongs to (M) , every descent datum on Y relative to $X \rightarrow X_1$ is effective; if Y_1 denotes the descended object, then $Y_1 \rightarrow X_1$ belongs to (N) .

Here is an example, to be treated later: $\mathcal{C}$ is the category of schemes, $\mathcal{T}$ is the faithfully flat quasi-compact topology, (M) is the family of faithfully flat quasi-compact morphisms, and (N) is the family of closed immersions, or the family of quasi-compact immersions.⁴⁷

Taking 4.6.11 into account, the principal result of 4.6.6 may be recalled as follows.

Proposition 4.7.1. *Let X be a squarable object of $\mathcal{C}$, and let R be an equivalence relation of type (M) in X . Suppose that $R \rightarrow X \times X$ belongs to (N) , where (N) satisfies (a) and $(f_{\mathcal{T}})$. Then the sheaf quotient X/R is given by*

$$(X/R)(S) = \{Z \subseteq X_S \mid Z \text{ is an } S\text{-subobject stable under } R \times S, \\ Z \rightarrow X_S \text{ belongs to } (N), \quad Z \rightarrow S \text{ is covering or belongs to } (M), \\ R_Z \simeq Z \times_S Z\}.$$

We also have the following result.

Proposition 4.7.2. *Let $X \in \text{Ob}(\mathcal{C})$, let R be an (M) -effective equivalence relation in X , and let (N) be a family of morphisms satisfying (a) and (f_M) . For every subobject Y of X which is stable under R and for which $Y \rightarrow X$ belongs to (N) , the equivalence relation R_Y induced in Y by R is (M) -effective. Moreover, the quotient*

$$Y/R_Y = Y'$$

⁴⁶Editor's note: Axiom (f_M) , which originally appeared before Proposition 4.7.2, has been moved here.

⁴⁷Editor's note: cf. §6.4; see also VI_A, 5.3.1.

is a subobject of $X' = X/R$, and $Y' \rightarrow X'$ belongs to (N) .

The map $Y \mapsto Y'$ is a bijection from the set of subobjects Y of X which are stable under R and for which $Y \rightarrow X$ belongs to (N) , onto the set of subobjects Y' of X' for which $Y' \rightarrow X'$ belongs to (N) . Its inverse is

$$Y' \longmapsto Y' \times_{X'} X.$$

Indeed, since R is (M) -effective, the morphism $X \rightarrow X'$ belongs to (M) . Let Y' be a subobject of X' such that the canonical morphism $Y' \rightarrow X'$ belongs to (N) . Then $Y = Y' \times_{X'} X$ is a subobject of X stable under R , and the morphisms $Y \rightarrow X$ and $Y \rightarrow Y'$ belong respectively to (N) and (M) , since both families are stable under base change. Let R_Y denote the equivalence relation induced in Y by R . By 4.4.11, the sheaf quotient Y/R_Y is represented by Y' ; hence 4.6.5 shows that R_Y is (M) -effective.

Conversely, let Y be a subobject of X , stable under R , such that $Y \rightarrow X$ belongs to (N) . Since Y is stable under R , its two inverse images in

$$R = X \times_{X'} X$$

are identical, and Y is equipped with a descent datum relative to $X \rightarrow X'$. The required conclusion follows from Axiom (f_M) for the family (N) .

Corollary 4.7.3. *Let $X \in \text{Ob}(\mathcal{C})$, and let R be an (M) -effective equivalence relation in X . Suppose in addition that $R \rightarrow X \times X$ belongs to (N) , where (N) satisfies (a) and (f_T) . Then, for every Y as in 4.7.2, $R_Y \rightarrow Y \times Y$ also belongs to (N) , and consequently 4.7.1 gives*

$$\begin{aligned} (Y/R_Y)(S) &= \{Z \subseteq Y_S \mid Z \text{ is an } S\text{-subobject stable under } R_Y \times S, \\ &\quad Z \rightarrow Y_S \text{ belongs to } (N) \text{ (hence so does } Z \rightarrow X_S), \\ &\quad Z \rightarrow S \text{ is covering, } R_Z \simeq Z \times_S Z\}. \end{aligned}$$

5. Passage to the Quotient and Algebraic Structures

5.1. Principal Homogeneous Bundles

Definition 5.1.0.⁴⁸ We recall (III 0.1) that an object X equipped with a right action of H is said to be *formally principal homogeneous*⁴⁹ under H if the canonical morphism of functors

$$X \times H \longrightarrow X \times X, \quad (x, h) \longmapsto (x, xh),$$

is an isomorphism. Equivalently (cf. *loc. cit.*), for every $S \in \text{Ob } \mathcal{C}$, the set $X(S)$ is formally principal homogeneous under $H(S)$: it is either empty or principal homogeneous under $H(S)$. In particular, when H acts on itself by right translations, it is formally principal homogeneous under itself.

Definition 5.1.1. An object X with an action of H is said to be *trivial* if it is isomorphic, as an H -object, to H with its translation action.

Proposition 5.1.2. *If X is formally principal homogeneous under H , there is an isomorphism*

$$\Gamma(X) \xrightarrow{\sim} \text{Isom}_{H\text{-obj.}}(H, X)$$

⁴⁸Editor's note: the numbering 5.1.0 has been introduced for later reference.

⁴⁹Editor's note: one also says "pseudo-torsor"; cf. EGA IV₄, 16.5.15. The more general notion of a formally homogeneous object (not necessarily principal homogeneous) is defined in Addendum 6.7.1 at the end of this exposé.

of principal homogeneous sets under $\Gamma(H)$.

To a section x of X , associate the morphism $H \rightarrow X$ defined on sets by $h \mapsto xh$. The assertion follows immediately by reduction to the set-theoretic case.

Corollary 5.1.3. *There is an isomorphism of H -objects*

$$X \xrightarrow{\sim} \text{Isom}_{H\text{-obj.}}(H, X).$$

Corollary 5.1.4. *An object with a group action is trivial if and only if it is formally principal homogeneous and has a section.*

Definition 5.1.5. Let $\mathcal{C}$ be a category equipped with a topology. An S -object X with an action of the S -group H is called a *principal homogeneous bundle under H* if it is locally trivial, that is, if the following equivalent conditions hold:⁵⁰

- (i) the collection of morphisms $T \rightarrow S$ for which the functor $X \times_S T$ is trivial under $H \times_S T$ is a refinement of S ;
- (ii) for a pretopology defining the given topology, there is a covering family $\{S_i \rightarrow S\}$ such that, for every i , the S_i -functor $X \times_S S_i$ is trivial under the S_i -group functor $H \times_S S_i$, equivalently, it has a section over S_i .

Proposition 5.1.6. *Let $\mathcal{C}$ be a category equipped with a topology $\mathcal{T}$, and let (M) be a family of morphisms of $\mathcal{C}$ satisfying axioms (a)–(e $\mathcal{T}$) of 4.6.3. Let H be an S -group whose structure morphism $H \rightarrow S$ belongs to (M) , and let P be an S -object with an action of H . The following conditions are equivalent:*

- (i) P is a principal homogeneous bundle under H (Definition 5.1.5);
- (ii) P is formally principal homogeneous under H , and its structure morphism $P \rightarrow S$ belongs to (M) ;
- (iii) there exists a morphism $S' \rightarrow S$ belonging to (M) such that, after base change from S to S' , P becomes trivial; that is, $P \times_S S'$ is trivial under $H \times_S S'$;
- (iv) H acts freely and (M) -effectively on P , and the quotient P/H is isomorphic to S .

First, (ii) and (iv) are equivalent. In either case $P \rightarrow S$ belongs to (M) , hence is squarable, which ensures that the fiber products $H \times_S P$ and $P \times_S P$ are representable. Condition (ii) plainly implies (iii): one may take $S' = P$, since formal principal homogeneity implies that $P \times_S P$ is trivial under $H \times_S P$ by 5.1.4, its diagonal section providing a section. Condition (iii) plainly implies (i), because $\{S' \rightarrow S\}$ is a covering family by axiom (d $\mathcal{T}$).

It remains to prove (i) $\Rightarrow$ (ii). The morphism of sheaves

$$P \times_S H \longrightarrow P \times_S P$$

is locally an isomorphism and hence is an isomorphism (4.5.8); thus P is formally principal homogeneous. The structure morphism $P \rightarrow S$ is locally isomorphic to the structure morphism $H \rightarrow S$, which belongs to (M) . It therefore belongs to (M) by axiom (e $\mathcal{T}$).

The equivalence between (i) and (iv) admits the following generalization.

Proposition 5.1.7. *Under the same assumptions on $\mathcal{C}$ and (M) , let H be an S -group and let X be an S -object on which H acts on the right. Suppose that $H \rightarrow S$ belongs to (M) . The following conditions are equivalent:*

⁵⁰Editor's note: in this case one also says that X is an H -torsor.

- (i) H acts freely and (M) -effectively on X ;
- (ii) there exists an S -morphism $p : X \rightarrow Y$, compatible with the equivalence relation on X defined by the action of H , such that the induced action of $H \times_S Y$ on X over Y makes X a principal homogeneous bundle under H_Y over Y .

Under these conditions, p identifies Y with the quotient X/H .

If $p : X \rightarrow Y$ is compatible with the action of H , then the induced action of $H \times_S Y$ on X over Y defines the same equivalence relation on X as the original action of H , by the isomorphism

$$H_Y \times_Y X \xrightarrow{\sim} H \times_S X.$$

The proposition follows from this observation and the preceding equivalence (iv) $\Leftrightarrow$ (i).

Corollary 5.1.7.1.⁵¹ *Let $\mathcal{C}$ be a category having a final object and fiber products, and let it be equipped with a topology $\mathcal{T}$ coarser than the canonical topology. Let $f : G \rightarrow H$ be a morphism of $\mathcal{C}$ -groups, put $K = \text{Ker}(f)$, and suppose that f is covering for $\mathcal{T}$. Then H represents the sheaf quotient G/K , and f is a K_H -torsor. Equivalently, one may say that G is a K -torsor over H .*

Indeed, because f is covering, it is a universal effective epimorphism (4.4.3). Hence, by 3.3.3.1, H is the quotient of G by the equivalence relation

$$\mathcal{R}(f) = G \times_H G.$$

Moreover, the morphism

$$G \times K \longrightarrow G \times_H G, \quad (g, k) \longmapsto (g, gk),$$

is an isomorphism of $K_G = G \times_H K_H$ -objects; its inverse is $(g, g') \mapsto (g, g^{-1}g')$. Thus $\mathcal{R}(f)$ is the equivalence relation defined by K . Since $f : G \rightarrow H$ is covering, it is a K_H -torsor by 5.1.6(ii), or directly by Definition 5.1.5(ii).

We can now sharpen Theorem 4.6.6 in the case of a quotient by a group action.

Proposition 5.1.8. *Under the assumptions of 5.1.7, let F_0 denote the functor over S defined as follows: for each $S' \rightarrow S$, $F_0(S')$ is the set of representable S' -subfunctors Z of $X \times_S S'$ that are stable under $H \times_S S'$ and are principal homogeneous bundles under this S' -group for the induced action (3.2.2).*

- (i) *The following conditions are equivalent:*

- (a) *the action of H on X is (M) -effective and free;*⁵²
- (b) *F_0 is representable.*

Under these conditions, $F_0 = X/H$.

- (ii) *Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S'_i \rightarrow S'\}$ and every family of morphisms $\{T_i \rightarrow S'_i\}$ in (N) , every descent datum on the T_i relative to $\{S'_i \rightarrow S'\}$ is effective. Suppose that $X \times_S H \rightarrow X \times_S X$ belongs to (N) , and that X is squarable. Then the element $p \in \text{Hom}(X, F_0)$ corresponding to the subobject $X \times_S H$ of $X \times_S X$ identifies F_0 with the sheaf quotient X/H .*

⁵¹Editor's note: this special case has been added as a corollary; it will be used several times in the following exposés.

⁵²Editor's note: the words "and free" have been added.

5.2. Group Structures and Passage to the Quotient

In this subsection we are concerned with the algebraic structures that can be placed on the quotient (G/H) of a group by a subgroup. We first work in the category of sheaves on $\mathcal{C}$ for an arbitrary topology. Taking the canonical topology and using 4.5.12 will then give results concerning universal effective passage to the quotient in $\mathcal{C}$.

Proposition 5.2.1. *Let $u : H \rightarrow G$ be a monomorphism of sheaves of groups. There is a unique G -object structure on the quotient sheaf G/H for which the canonical morphism*

$$p : G \longrightarrow G/H$$

is a morphism of G -objects. This structure is functorial in the pair (G, H) : given a commutative diagram

$$\begin{array}{ccc} H & \longrightarrow & G \\ \downarrow & & \downarrow f \\ H' & \longrightarrow & G', \end{array}$$

the morphism $f : G/H \rightarrow G'/H'$ of 3.2.3 is compatible with the morphism f between the acting groups.

Indeed, G/H is the sheaf associated with the presheaf

$$i(G)/i(H) : S \longmapsto G(S)/H(S).$$

Since the functor a is left exact, it carries objects with a group action to objects with a group action. The presheaf $i(G)/i(H)$ is an $i(G)$ -object, and therefore

$$G/H = a(i(G)/i(H))$$

is an $a(i(G)) = G$ -object. This construction plainly has all the stated properties.

Corollary 5.2.2. *Let $u : H \rightarrow G$ be a monomorphism of $\mathcal{C}$ -groups, and suppose that the action of H on G is universally effective. There is a unique G -object structure on the quotient object $G/H \in \text{Ob } \mathcal{C}$ for which $p : G \rightarrow G/H$ is a morphism of objects with an action. In the sense of the preceding proposition, this structure is functorial in the pair (H, G) , with H acting universally effectively on G .*

Proposition 5.2.3. *Let $u : H \rightarrow G$ be a monomorphism of sheaves of groups that identifies H with an invariant subgroup sheaf of G . There is a unique sheaf-of-groups structure on the quotient sheaf G/H for which the canonical morphism $p : G \rightarrow G/H$ is a group morphism. This structure is functorial in the pair (H, G) , with H invariant.*

The proof is similar to that of 5.2.1.

Corollary 5.2.4. *Let $u : H \rightarrow G$ be a monomorphism of $\mathcal{C}$ -groups that identifies H with an invariant subgroup of G . Suppose that the action of H on G is universally effective. There is a unique group structure on the quotient object $G/H \in \text{Ob } \mathcal{C}$ for which the canonical morphism $G \rightarrow G/H$ is a group morphism. This structure is functorial in the pair (H, G) , with H invariant and acting universally effectively.*

The group structure of G/H can be characterized more explicitly.

Proposition 5.2.5. *Under the assumptions of 5.2.4, let K be a $\mathcal{C}$ -group and let $f : G \rightarrow K$ be a morphism. The following conditions are equivalent:*

- (i) *f is a group morphism compatible with the equivalence relation defined by H ;*
- (ii) *f is a group morphism whose restriction $H \rightarrow K$ is the trivial morphism;*

(iii) f factors through a group morphism $G/H \rightarrow K$.

In particular, there is an isomorphism, functorial in the group K ,

$$\mathrm{Hom}_{\mathcal{C}\text{-gr.}}(G/H, K) \xrightarrow{\sim} \{f \in \mathrm{Hom}_{\mathcal{C}\text{-gr.}}(G, K) \mid f \circ u = e\}.$$

The equivalence of (i) and (ii) is proved set-theoretically, and (iii) $\Rightarrow$ (ii) is immediate. The equivalence of (iii) and (ii) follows from the formula

$$\mathrm{Hom}(G/H, K) \simeq \mathrm{Hom}(i(G)/i(H), K)$$

and the definition of the group structure on G/H .

Remark 5.2.6. In the preceding situation, if the kernel of f is exactly H , then the morphism $G/H \rightarrow K$ through which f factors is a monomorphism. This follows at once from 3.3.4.

For sheaves of groups, 4.4.11 can be sharpened as follows.

Proposition 5.2.7. *Let G be a sheaf of groups and H an invariant subgroup sheaf. For every subgroup sheaf K of G containing H , let $K' = K/H$, regarded as a subgroup of $G' = G/H$. Then*

$$K = K' \times_{G'} G,$$

and the maps

$$K \longmapsto K/H, \quad K' \longmapsto K' \times_{G'} G$$

give a bijection between the subgroup sheaves of G containing H and the subgroup sheaves of G' . Under this correspondence, the invariant subgroup sheaves of G and of G' correspond.

The first assertion follows readily from 4.4.11 and 3.2.4. It remains to show that K is invariant in G if and only if K' is invariant in G' . If K is invariant in G , then the presheaf $i(K)/i(H)$ is invariant in $i(G)/i(H)$; by the usual argument, the same is true of the associated sheaves. Conversely, if K' is invariant in G' , then the fiber product $K' \times_{G'} G$ is invariant in G , as is immediate.

Now let L be an arbitrary subgroup sheaf of G . Denote by $\bar{L}$ the saturation of L for the equivalence relation defined by H ; we also write $\bar{L} = L \cdot H$.

Proposition 5.2.8. *Under the preceding assumptions, $L \cdot H$ is a subgroup sheaf of G containing H , and the image of L in G/H is identified with*

$$(L \cdot H)/H \simeq L/(H \cap L).$$

Indeed, let L' denote the image sheaf of L in G/H . Under the correspondence of the preceding proposition, it is the subgroup sheaf of G/H corresponding to $L \cdot H$. Since $L \rightarrow L'$ is covering, and hence a universal effective epimorphism of sheaves, it follows from 4.4.9 that L' is identified with the quotient of L by the kernel of $L \rightarrow L'$, which is plainly $H \cap L$.

Finally, consider a sheaf of groups G , a subgroup sheaf K , and an invariant subgroup sheaf H of K . We first define a right action of the sheaf of groups $H \backslash K (= K/H)$ on G/H . The group K acts on G by right translations. Since H is invariant in K , this action is compatible with the equivalence relation defined by the action of H , and therefore induces an action of K on G/H , that is, a morphism

$$K^\circ \longrightarrow \mathrm{Aut}(G/H)$$

from the group opposite to K . Since the target is a sheaf (4.5.13) and this morphism is trivial on H , it factors through K/H and defines the desired action. The right and left actions of G on itself commute; hence the actions of G and K/H on G/H commute.

Proposition 5.2.9. *Under the preceding assumptions, K/H acts freely on the right on G/H , and there is a canonical isomorphism of G -objects*

$$(G/H)/(K/H) \simeq G/K.$$

If K is invariant in G , in which case K/H is invariant in G/H by 5.2.7, this isomorphism respects the group structures on both sides.

There is an isomorphism of presheaves

$$(i(G)/i(H))/(i(K)/i(H)) \xleftarrow{\sim} i(G)/i(K),$$

which respects the $i(G)$ -object structures. Applying the functor a to this relation gives the asserted result.

Corollary 5.2.10. *Let G be a $\mathcal{C}$ -group, K a $\mathcal{C}$ -subgroup of G , and H an invariant $\mathcal{C}$ -subgroup of K . Let (M) be a family of morphisms of $\mathcal{C}$ satisfying axioms (a)–(e $\mathcal{T}$). Suppose that the right action of H on G (respectively, on K) is (M) -effective. Then K/H acts naturally and freely on the right on G/H , and this action commutes with that of G . The following conditions are equivalent:*

- (i) *the action of K on G is (M) -effective;*
- (ii) *the action of K/H on G/H is (M) -effective.*

Under these conditions, there is an isomorphism of objects⁵³ with a G -action

$$(G/H)/(K/H) \simeq G/K.$$

5.3. Use of Effectivity Criteria: Noether's Theorem

⁵⁴ Let $\mathcal{C}$, $\mathcal{T}$, and (M) be as usual. Let (N) be a family of morphisms satisfying axioms (a) and (f_M) of 4.7. Combining 5.2.7 and 4.7.2 gives the following result.

Proposition 5.3.1. *Let G be a $\mathcal{C}$ -group, and let H be an invariant $\mathcal{C}$ -subgroup of G that acts (M) -effectively on G . For every $\mathcal{C}$ -subgroup K of G containing H such that $K \rightarrow G$ belongs to (N) , the action of H on K is (M) -effective, and the quotient $K/H = K'$ is a $\mathcal{C}$ -subgroup of $G/H = G'$ such that $K' \rightarrow G'$ belongs to (N) . The map $K \mapsto K'$ is a bijection between the set of $\mathcal{C}$ -subgroups K of G that contain H and for which $K \rightarrow G$ belongs to (N) , and the set of $\mathcal{C}$ -subgroups K' of G' for which $K' \rightarrow G'$ belongs to (N) . Its inverse is*

$$K' \longmapsto K \times_{G'} G.$$

Under this correspondence, the invariant subgroups of G and of G' correspond.

Corollary 5.3.2. *If $H \rightarrow G$ belongs to (N) , then $\mathcal{C}$ has a final object e , and the unit section $e \rightarrow G/H$ belongs to (N) .⁵⁵*

⁵³Editor's note: of $\mathcal{C}$.

⁵⁴Editor's note: in fact, the isomorphisms established in 5.2.8–5.2.10 deserve to be called “Noether isomorphism theorems.”

⁵⁵Editor's note: this follows by applying the proposition with $K = H$.

6. Topologies in the Category of Schemes

6.1. The Zariski Topology

This is the topology generated by the following pretopology: a family of morphisms $\{S_i \rightarrow S\}$ is covering if each morphism is an open immersion and the union of the images of the S_i is all of S . It is denoted by (Zar).

Definition 6.1.1. *A sheaf for the Zariski topology is also called a functor of local nature: it is a contravariant functor from **Sch** to **Set** such that, for every scheme S and every covering of S by open subschemes S_i , the diagram*

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \cap S_j)$$

is exact.

In particular, a functor of local nature carries direct sums to products. Since every representable functor is a sheaf, the Zariski topology is coarser than the canonical topology.

As a matter of terminology, whenever we say "local" or "locally" without further qualification, the reference is to the Zariski topology and hence to the usual meaning of these words.

6.2. A Procedure for Constructing Topologies

Proposition 6.2.1. *Let $\mathcal{C}$ be a category, let $\mathcal{C}'$ be a full subcategory, and let P be a set of families of morphisms in $\mathcal{C}$ with common target, stable under base change and composition (that is, satisfying axioms (P 1) and (P 2) of 4.2.5). Let P' be a set of families of morphisms in $\mathcal{C}'$ that contains every singleton family consisting of an identity isomorphism. Equip $\mathcal{C}$ with the topology generated by P and P' (cf. 4.2.5.0), and suppose that the following three conditions hold:*

- (a) *If $\{S_i \rightarrow S\} \in P'$, so that $S_i, S \in \text{Ob } \mathcal{C}'$, and $T \rightarrow S$ is a morphism in $\mathcal{C}'$, then the fiber products $S_i \times_S T$, formed in $\mathcal{C}$, exist and the family $\{S_i \times_S T \rightarrow T\}$ belongs to P' ; in particular, $S_i \times_S T \in \text{Ob } \mathcal{C}'$.*

This condition implies that P' is stable under base change in $\mathcal{C}'$, but is not equivalent to that assertion: it also requires the inclusion functor $\mathcal{C}' \hookrightarrow \mathcal{C}$ to commute with certain fiber products.

- (b) *For every $S \in \text{Ob } \mathcal{C}$, there exists a family $\{S_i \rightarrow S\} \in P$ with $S_i \in \text{Ob } \mathcal{C}'$ for every i .*

- (c) *In a situation*

$$\begin{array}{ccc} & & S_{ijk} \\ & & \downarrow (P') \\ S_i & \xleftarrow{(P)} & S_{ij} \\ \downarrow (P') & & \\ S & & \end{array} \quad (6.2.1.c)$$

where S, S_i, S_{ij}, S_{ijk} are objects of $\mathcal{C}'$, where $\{S_i \rightarrow S\} \in P'$, $\{S_{ij} \rightarrow S_i\} \in P$ for each i , and $\{S_{ijk} \rightarrow S_{ij}\} \in P'$ for each pair ij , there exists a family $\{T_n \rightarrow S\} \in P'$, and for each n a multi-index (ijk) , fitting into a commutative diagram

$$\begin{array}{ccc} S_{ijk} & \xleftarrow{\quad} & T_n \\ & \searrow & \downarrow \\ & & S. \end{array} \quad (6.2.1.d)$$

Under these hypotheses, a sieve R of an object $S \in \text{Ob } \mathcal{C}$ is covering if and only if there is a composite family

$$\begin{array}{ccc} R & \leftarrow & S_{ij} \\ \downarrow & & \downarrow (P') \\ S & \xleftarrow{(P)} & S_i \end{array} \quad (6.2.1.e)$$

where $S_i, S_{ij} \in \text{Ob } \mathcal{C}'$, $\{S_i \rightarrow S\} \in P$, and $\{S_{ij} \rightarrow S_i\} \in P'$ for every i , such that the resulting morphisms $S_{ij} \rightarrow S$ factor through R . Equivalently, the sieve generated by this composite family is contained in R .

Proof. Families belonging to P and P' are covering, and therefore so is a composite of such families, by (C 2). A sieve of the displayed form is thus covering because it contains a covering sieve.

Conversely, it is enough to show that the sieves of this form constitute a topology, that is, to verify axioms (T 1)–(T 4) of 4.2.1.

Axiom (T 4). Let $S \in \text{Ob } \mathcal{C}$. By (b), there exists a family $\{S_i \rightarrow S\} \in P$ with $S_i \in \text{Ob } \mathcal{C}'$. The singleton families $\{S_i \xrightarrow{\text{id}_{S_i}} S_i\}$ belong to P' by hypothesis. Hence the maximal sieve S of S has the required form:

$$\begin{array}{ccc} S_i & \longrightarrow & S \\ (P') \downarrow \text{id} & & \downarrow \text{id}_S \\ S_i & \xrightarrow{(P)} & S. \end{array} \quad (6.2.1.T4)$$

Axiom (T 3). This is evident.

Axiom (T 2). Let R be a sieve of S of the required form, and let C be a sieve⁵⁶ of S such that, for every $T \in \text{Ob } \mathcal{C}$ and every morphism $T \rightarrow S$ factoring through R , the sieve $C \times_S T$ of T has the required form. Since $S_{ij} \rightarrow S$ factors through R , the sieve

$$C_{ij} = C \times_S S_{ij}$$

of S_{ij} has the required form; it is represented schematically by

$$\begin{array}{ccccc} S_{ij} & \longleftarrow & C_{ij} & & \\ (P') \downarrow & & \downarrow & & \\ S_i & & C & & \\ (P) \downarrow & \swarrow & & & \\ R & \longrightarrow & S & \longleftarrow & C. \end{array} \quad (6.2.1.T2a)$$

Consequently, for every pair ij , there is a diagram of the form

$$\begin{array}{ccc} S_{ijkl} & & \\ (P') \downarrow & \searrow & \\ S_{ijk} & & \\ (P) \downarrow & & \\ S_{ij} & \longleftarrow & C_{ij}. \end{array} \quad (6.2.1.T2b)$$

We have therefore produced a composite family

$$S_{ijkl} \xrightarrow{(P')} S_{ijk} \xrightarrow{(P)} S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S \quad (6.2.1.T2c)$$

⁵⁶Editor's note: the original has been corrected here.

belonging to $P \circ P' \circ P \circ P'$, factoring through C , and having every object other than S in $\mathcal{C}'$.

Apply condition (c) to each family $\{S_{ijkl} \rightarrow S_i\}$. For every i we obtain a family $\{T_{in} \rightarrow S_i\} \in P'$ such that $T_{in} \rightarrow S$ factors through one of the S_{ijkl} , and hence through C :

$$\begin{array}{ccc} T_{in} & \longrightarrow & S_{ijkl} \\ (P') \downarrow & & \downarrow C \\ S_i & & \\ (P) \downarrow & \swarrow & \\ S & & \end{array} \quad (6.2.1.T2d)$$

Thus the sieve C of S has the required form, completing the verification of (T 2).

Axiom (T 1). Let R be a sieve of S of the prescribed form and let $T \rightarrow S$ be a morphism in $\mathcal{C}$. We show that the sieve $R \times_S T$ of T again has the prescribed form. Consider the commutative configuration

$$\begin{array}{ccccccc} S_{ij} & \longleftarrow & & & U_{ikj} & & \\ (P') \downarrow & & & & (P') \downarrow & & \\ S_i & \longleftarrow & T_i & \xleftarrow{(P)} & U_{ik} & & \\ (P) \downarrow & & (P) \downarrow & & \swarrow (P) & & \\ R & \longrightarrow & S & \longleftarrow & T. & & \end{array} \quad (6.2.1.T1a)$$

Put $T_i = S_i \times_S T$. The family $\{T_i \rightarrow T\}$ belongs to P , by (P 1). By (b), choose families $\{U_{ik} \rightarrow T_i\} \in P$ with $U_{ik} \in \text{Ob } \mathcal{C}'$. By (P 2), $\{U_{ik} \rightarrow T\} \in P$. Condition (a) shows that

$$U_{ik} \times_{S_i} S_{ij} = U_{ikj}$$

is an object of $\mathcal{C}'$ and that, for each ik , $\{U_{ikj} \rightarrow U_{ik}\} \in P'$. The commutative diagram

$$\begin{array}{ccccc} R & \longleftarrow & U_{ikj} & & \\ \downarrow & & (P') \downarrow & & \\ S & \longleftarrow & T & \xleftarrow{(P)} & U_{ik} \end{array} \quad (6.2.1.T1b)$$

shows that the morphisms $U_{ikj} \rightarrow T$ factor through the sieve $T \times_S R$ of T . This sieve therefore has the required form, and the proof is complete.

Corollary 6.2.2. *Let $S \in \text{Ob } \mathcal{C}'$, and let R be a sieve of S . Then R is covering if and only if there exists a family $\{T_i \rightarrow S\} \in P'$ that factors through R .*

Such a sieve is plainly covering. Conversely, apply condition (c) with the family $\{S_i \rightarrow S\}$ reduced to the identity isomorphism of S ; the proposition then shows that every covering sieve has the stated form.

Corollary 6.2.3. *A presheaf F on $\mathcal{C}$ is separated (respectively, is a sheaf) if and only if the map*

$$F(S) \longrightarrow \prod_i F(S_i)$$

is injective (respectively, the diagram

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times_S S_j)$$

is exact) in each of the following two cases:

- (i) $\{S_i \rightarrow S\} \in P$;
- (ii) $S, S_i \in \text{Ob } \mathcal{C}'$ and $\{S_i \rightarrow S\} \in P'$.

The conditions are necessary because the families in question are covering. Let C be the sieve of S that is the image of a composite family

$$\{S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S\}.$$

A straightforward diagram chase shows that the conditions of the corollary imply that

$$\text{Hom}(S, F) \xrightarrow{g} \text{Hom}(C, F)$$

is injective, respectively bijective. Every covering sieve R of S contains a sieve C of this form, and there is a commutative diagram

$$\begin{array}{ccc} \text{Hom}(S, F) & \xrightarrow{f} & \text{Hom}(R, F) \\ & \searrow g & \downarrow h \\ & & \text{Hom}(C, F). \end{array} \quad (6.2.3.1)$$

Since g is injective, so is f , and hence F is separated. Moreover, R refines C , so h is also injective. If g is bijective, then f is bijective as well, and F is a sheaf.

Remark 6.2.4. *The preceding corollary does not follow from 4.3.5, since P' is not stable under base extension.*

Remark 6.2.5. *Condition (c) holds in particular when:*

- (i) P' is stable under composition;
- (ii) whenever $\{S_j \rightarrow S\}$ is a family of morphisms in $\mathcal{C}'$ belonging to P , it contains a subfamily belonging to P' .

6.3. Application to the Category of Schemes

Take $\mathcal{C}$ to be the category of schemes, $\mathcal{C}'$ the full subcategory of affine schemes, and P the set of surjective families of open immersions. We shall consider the following sets P'_i :

- P'_1 : finite surjective families of flat morphisms;
- P'_2 : finite surjective families of flat, finitely presented, quasi-finite morphisms;
- P'_3 : finite surjective families of étale morphisms;
- P'_4 : finite surjective families of finite étale morphisms.

The description of P'_2 is accompanied by the following clarification.⁵⁷

For each P'_i , except P'_4 , the conditions of Proposition 6.2.1 hold. Condition (c) follows from 6.2.5: since an affine scheme is quasi-compact, every family of morphisms in $\mathcal{C}'$ belonging to P contains a finite subfamily that still belongs to P , and hence belongs to P'_i for

⁵⁷Editor's note: let P''_2 be the set of finite surjective families of morphisms in $\mathcal{C}'$ that are flat and of finite presentation. By Proposition 6.3.1 below, the topology T_2 generated by P and P'_2 coincides with the topology generated by P and P''_2 . This follows from the results of EGA IV₄, §17.16 on quasi-sections; see the proof of 6.3.1. We recall the following special case of EGA IV₄, 17.16.2. Let S be affine and let $f : X \rightarrow S$ be a surjective flat morphism locally of finite presentation. Then there exist a faithfully flat, finitely presented, quasi-finite morphism $S' \rightarrow S$, with S' affine, and an S -morphism $S' \rightarrow X$.

$i = 1, 2, 3$. The topology T_i generated by P and P'_i is denoted and named as follows:⁵⁷

- $T_1 = (\text{fpqc})$ the faithfully flat quasi-compact topology,
- $T_2 = (\text{fppf})$ the faithfully flat topology (locally of finite presentation),
- $T_3 = (\text{ét})$ the étale topology,
- $T_4 = (\text{étf})$ the finite étale topology.

Since $P'_1 \supset P'_2 \supset P'_3 \supset P'_4$, one has

$$(\text{fpqc}) \geq (\text{fppf}) \geq (\text{ét}) \geq (\text{étf}) \geq (\text{Zar}).$$

Proposition 6.3.1.

- (i) A sieve R of S is covering for T_i , $1 \leq i \leq 3$, if and only if there exist a covering (S_p) of S by affine open subschemes and, for every p , a family $\{S_{pq} \rightarrow S_p\} \in P'_i$, with all S_{pq} affine, such that every morphism $S_{pq} \rightarrow S$ factors through R .⁵⁸
- (ii) A presheaf F on **Sch** is a sheaf for (fpqc) (respectively, (fppf) , (ét) , or (étf)) if and only if:
 - (a) F is a sheaf for (Zar) , that is, a functor of local nature;
 - (b) for every faithfully flat (respectively, faithfully flat, finitely presented and quasi-finite; surjective étale; or surjective finite étale) morphism $T \rightarrow S$, with T and S affine, the diagram

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact.

- (iii) The topologies T_i , for $i = 1, 2, 3, 4$, are coarser than the canonical topology.
- (iv) Every surjective family of flat open morphisms (respectively, flat morphisms locally of finite presentation; étale morphisms; or finite étale morphisms) is covering for (fpqc) (respectively, (fppf) , (ét) , or (étf)).
- (v) Every finite surjective family of flat quasi-compact morphisms is covering for (fpqc) . In particular, every faithfully flat quasi-compact morphism is covering for (fpqc) .

Proof. Part (i) follows from 6.2.1 and part (ii) from 6.2.3, taking into account that a sheaf for the Zariski topology carries direct sums to products. Every representable functor is a sheaf for (Zar) and satisfies condition (ii)(b), by SGA 1, VIII, 5.3. Thus T_1 is coarser than the canonical topology, which proves (iii).

We prove (iv). Let $\{S_i \rightarrow S\}$ be a family as in the statement. By covering S with affine open subschemes, we reduce immediately to the case in which S is affine.⁵⁹

First suppose that the morphisms $S_i \rightarrow S$ are flat and open (respectively, étale). Cover each S_i by affine open subschemes S_{ij} . Since the morphisms are open, the images T_{ij} of the S_{ij} form an open covering of S . The affine scheme S is quasi-compact, so finitely many T_{ij} , indexed by a finite set E , cover it. Then

$$S' = \coprod_{(i,j) \in E} S_{ij}$$

⁵⁸Editor's note: every family $\{S_{pq} \rightarrow S_p\} \in P'_i$ is finite. Hence $S'_p = \coprod_q S_{pq}$ is affine, and the family may be replaced by the morphism $S'_p \rightarrow S_p$, which still belongs to P'_i .

⁵⁹Editor's note: the argument that follows has been simplified by using the fact that S is now assumed affine.

is affine, and $S' \rightarrow S$ belongs to P'_1 , respectively P'_3 , and is therefore covering. Since it factors through the given family, that family is covering.

In the (étf) case, every S_i is finite over S and hence affine. In the preceding argument one may take the singleton covering $\{S_i\}$ of S_i , which gives $S' \rightarrow S \in P'_4$.

Now suppose that the morphisms $f_i : S_i \rightarrow S$ are flat and locally of finite presentation. For every $s \in S$, the proof of EGA IV₄, 17.16.2 supplies an affine subscheme $X(s)$ of some S_i , with $s \in f_i(X(s))$, such that the restriction

$$g_i : X(s) \longrightarrow S$$

is flat, finitely presented, and quasi-finite. Its image $g_i(X(s))$ is an open neighborhood $U(s)$ of s , by EGA IV₂, 2.4.6. Since S is affine, finitely many such opens $U(s_j)$, $j = 1, \dots, n$, cover S . Consequently,

$$X' = \coprod_{j=1}^n X(s_j)$$

is affine and $X' \rightarrow S$ is surjective, flat, finitely presented, and quasi-finite, so it belongs to P'_2 .⁶⁰ This completes the proof of (iv).

We prove (v). Let $\{S_i \rightarrow S\}$ be a finite surjective family of flat quasi-compact morphisms, and cover S by affine open subschemes T_j . The schemes

$$S_{ij} = T_j \times_S S_i$$

are quasi-compact, and therefore admit finite affine open coverings T_{ijk} . Each $T_{ijk} \rightarrow T_j$ is flat, and the family $\{T_{ijk} \rightarrow T_j\}$ is finite and surjective, hence covering for T_j . By composition, $\{T_{ijk} \rightarrow S\}$ is covering as well. It factors through the given family, which is therefore covering:

$$\begin{array}{ccccc} S_i & \longleftarrow & S_{ij} & \longleftarrow & T_{ijk} \\ \downarrow & & \downarrow & \swarrow & \\ S & \longleftarrow & T_j & & \end{array} \quad (6.3.1.v)$$

Corollary 6.3.2. *Let (M_i) be the following families of morphisms:*

- (M_1) : faithfully flat quasi-compact morphisms;
- (M_2) : faithfully flat morphisms locally of finite presentation;
- (M_3) : surjective étale morphisms;
- (M_4) : surjective finite étale morphisms.

*The family (M_i) satisfies axioms (a), (b), (c), (d_{T_i}) , and (e_{T_i}) of 4.6.3.*⁶¹

Conditions (a), (b), and (c) are classical (see EGA and SGA 1, passim).⁶² By 6.3.1(iv),(v), (M_i) satisfies (d_{T_i}) . It remains to verify (e_{T_i}) . It is enough to verify (e_{T_1}) , which implies all the others; this follows from SGA 1, VIII, §§4–5.

Corollary 6.3.3. *Let X be a scheme and let R be an equivalence relation in X of type (M_i) . Then R is (M_i) -effective if and only if the sheaf quotient of X by R for T_i is representable; in that case it is represented by the quotient X/R .*

This is 4.6.5.

⁶⁰Editor's note: this shows that if P'_2 denotes the set of finite surjective families of flat, finitely presented morphisms in $\mathcal{C}'$, then the topology generated by P and P'_2 equals T_2 . Moreover, in the notation used at the beginning of the proof of (iv), if each S_i is covered by affine opens of finite presentation over S , then the resulting morphism $S' \rightarrow S$ belongs to P'_2 .

⁶¹Editor's note: the original has been corrected by adding surjectivity to (M_3) and (M_4) ; it is automatically satisfied in the other two cases.

⁶²Editor's note: see EGA I, 6.6.4 for "quasi-compact"; EGA II, 6.1.5 for "finite"; EGA IV₁, 1.6.2 for "locally of finite presentation"; EGA IV₂, 2.2.13 for "faithfully flat"; and EGA IV₄, 17.3.3 for "étale".

6.4. Effectivity Conditions

We now seek families (N) of morphisms satisfying axiom (f_T) of 4.7. Observe first that (f_{T_1}) implies (f_{T_i}) , so it is enough to consider the (fpqc) topology.

Lemma 6.4.1. *The following families of morphisms satisfy axiom (f_{T_1}) of 4.7, that is, they "descend by (fpqc)":*

- (N) : open immersions;
- (N') : closed immersions;
- (N'') : quasi-compact immersions.

By 6.3.1(ii), it is enough to verify that these families descend for the Zariski topology and along a faithfully flat quasi-compact morphism. The first assertion is clear. For (N) , the second is SGA 1, VIII, 4.4; for (N') , it is loc. cit., 1.9; for (N'') , argue as in loc. cit., 5.5, using the two preceding results.

Corollary 6.4.2. *The same result holds for quasi-compact open immersions.*

These results make the general results of 4.7.1, 4.7.2, 5.1.8, 5.3.1, and so on applicable to the present situation. We state the first as an example.

Corollary 6.4.3. (= 4.7.1 + 4.6.10). *Let X be a scheme and R an equivalence relation in X . Suppose that $R \rightarrow X$ is faithfully flat and quasi-compact, and that $R \rightarrow X \times X$ is a closed immersion (respectively, open, quasi-compact, or quasi-compact open). Then the sheaf quotient X/R is the same for the (fpqc) topology as for the canonical topology. For every scheme S ,*

$$(X/R)(S) = \left\{ \begin{array}{l} \text{closed (respectively, open, retrocompact, or open retrocompact)} \\ \text{subschemes } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ such that} \\ Z \rightarrow S \text{ is faithfully flat and quasi-compact and} \\ R_Z \rightrightarrows Z \rightarrow S \text{ is exact} \end{array} \right\}. \quad (6.4.3.1)$$

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6.5. Principal Homogeneous Bundles

We merely record the terminology:

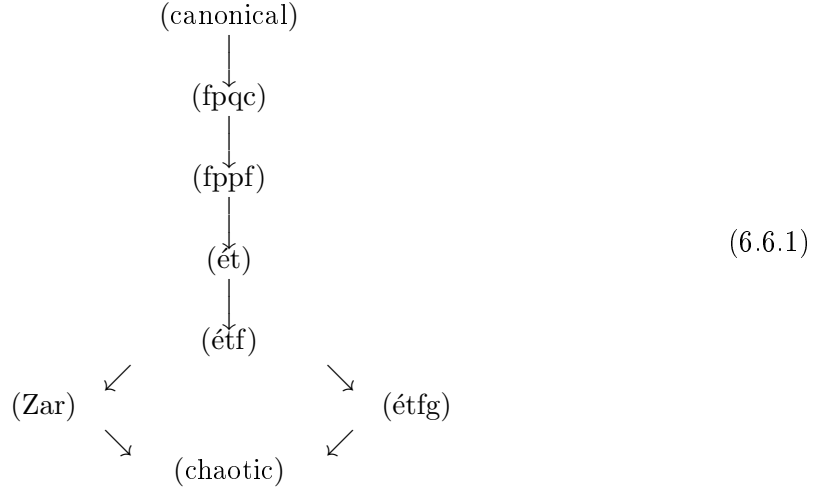
topology	principal homogeneous bundles	
(fpqc)	principal homogeneous bundles (without qualifier)	(6.5.1)
(ét)	quasi-isotrivial principal homogeneous bundles	
(étf)	locally isotrivial principal homogeneous bundles	
(Zar)	locally trivial principal homogeneous bundles.	

6.6. Other Topologies

Other topologies on the category of schemes are sometimes useful. One is the *global finite étale topology* (étfg), generated by the pretopology whose covering families are the surjective families of finite étale morphisms. It is not finer than the Zariski topology. The corresponding principal homogeneous bundles are called *isotrivial*.

⁶³Editor's note: recall that a subscheme Z of a scheme T is called *retrocompact* if the immersion $Z \hookrightarrow T$ is quasi-compact; see EGA 0_{III}, 9.1.1.

The position of this topology, and of the chaotic topology, in the lattice is as follows:



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6.7. Homogeneous Spaces

Let⁶⁵ G be an S -group scheme, let X be an S -scheme equipped with a left action of G , and let

$$\Phi : G \times_S X \longrightarrow X \times_S X$$

be the morphism of S -schemes defined on points by

$$(g, x) \longmapsto (gx, x).$$

Recall (cf. 5.1.0 and III.0.1) that X is called a *formally principal homogeneous space* under G if the following equivalent conditions hold:

- (i) for every $T \rightarrow S$, the set $X(T)$ is empty or is a principal homogeneous space under $G(T)$;
- (ii) Φ is an isomorphism of S -functors;
- (iii) Φ is an isomorphism of S -schemes.

The equivalence (i) $\iff$ (ii) is clear, and (ii) $\iff$ (iii) holds because $\mathcal{C} = (\mathbf{Sch}/S)$ is a full subcategory of $\widehat{\mathcal{C}}$.

The notion of a *formally homogeneous space*, not necessarily principal homogeneous, is obtained by requiring Φ to be an epimorphism in the category of sheaves for an appropriate topology $\mathcal{T}$. Indeed, requiring Φ to be an epimorphism of S -functors is equivalent to requiring that, for every $T \rightarrow S$, the set $X(T)$ be empty or homogeneous (not necessarily principal homogeneous) under $G(T)$. This is too restrictive, as the following simple example shows.

Let

$$S = \operatorname{Spec} R, \quad G = \mathbf{G}_{m,R}, \quad X = \mathbf{G}_{m,R},$$

and let G act on X by $t \cdot x = t^2x$. Then Φ is surjective, finite, and étale, and hence is an epimorphism in the category of sheaves for the (étf) topology (and, *a fortiori*, an epimorphism

⁶⁴Editor's note: recall (cf. 4.4.2) that the chaotic topology is the coarsest topology, defined by $J(S) = \{S\}$ for every $S \in \operatorname{Ob} \mathcal{C}$.

⁶⁵Editor's note: the numbering that follows has been added.

of S -schemes). On the other hand, the points 1 and -1 of $X(R)$ are not conjugate under any element of $G(R)$. Thus

$$G(R) \times X(R) \longrightarrow X(R) \times X(R)$$

is not surjective.⁶⁶ We are therefore led to the following definition.

Definition 6.7.1. *Let G be an S -group, let X be an S -scheme equipped with an action of G , and let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. The scheme X is called a formally homogeneous space under G , relative to $\mathcal{T}$, if the following equivalent conditions hold:*

- (i) *the morphism $\Phi : G \times_S X \rightarrow X \times_S X$ is an epimorphism in the category of sheaves for $\mathcal{T}$;*
- (ii) *for every $T \rightarrow S$ and every $x, y \in X(T)$, there exist a $\mathcal{T}$ -covering morphism $T' \rightarrow T$ and an element $g \in G(T')$ such that $y_{T'} = g \cdot x_{T'}$.*

Remark 6.7.2. *Condition (i) implies in particular that Φ is a universal effective epimorphism in $(\mathbf{Sch}/S)$, by 4.4.3. As is readily seen, it follows that Φ is surjective; cf. 1.3, Editor's note (3).*

Proposition and Definition 6.7.3.⁶⁷ *Let G be an S -group, let X be an S -scheme equipped with an action of G , and let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. The following conditions are equivalent:*

- (i) *X satisfies both of the following conditions:*
 - (1) *the morphism $\Phi : G \times_S X \rightarrow X \times_S X$ is covering, that is, X is a formally homogeneous G -space;*
 - (2) *the structural morphism $X \rightarrow S$ is also covering, that is, locally for $\mathcal{T}$ it has a section (cf. 4.4.8 bis, (b)).*
- (ii) *Locally on S for the topology $\mathcal{T}$, the scheme X , with its G -action, is isomorphic to the quotient sheaf of G by a subgroup scheme H , for $\mathcal{T}$. In other words, there is a covering family $\{S_i \rightarrow S\}$ such that every $X \times_S S_i$ represents the sheaf quotient of $G \times_S S_i$ by some subgroup scheme H_i .*

Under these conditions, X is called a G -homogeneous space, relative to $\mathcal{T}$.

Proof. Assume (ii). Put

$$G_i = G \times_S S_i, \quad X_i = X \times_S S_i.$$

The scheme X_i has a section over S_i , namely the composite of the unit section $\varepsilon_i : S_i \rightarrow G_i$ with the projection $\pi_i : G_i \rightarrow X_i = G_i/H_i$. It follows that $X \rightarrow S$ is covering.

⁶⁶Editor's note: this difficulty plainly comes from the following fact. Let $\mathcal{C}'$ be a full subcategory of $\widehat{\mathcal{C}}$ containing $\mathcal{C}$, for example the category $\widehat{\mathcal{C}}_{\mathcal{T}}$ of sheaves on $\mathcal{C}$ for a topology $\mathcal{T}$ coarser than the canonical topology. If $f : X \rightarrow Y$ is a morphism in $\mathcal{C}$, then the implications

$$f \text{ epimorphic in } \widehat{\mathcal{C}} \implies f \text{ epimorphic in } \mathcal{C}' \implies f \text{ epimorphic in } \mathcal{C}$$

are generally strict.

⁶⁷Editor's note: cf. [Ray70], Definition VI.1.1.

Moreover, π_i is covering, so $\pi_i \times \pi_i$ is covering as well, by 4.2.3 (C 1),(C 2). There is a commutative diagram

$$\begin{array}{ccc} G_i \times_{S_i} X_i & \xrightarrow{\Phi_i} & X_i \times_{S_i} X_i \\ \text{id} \times \pi_i \uparrow & & \pi_i \times \pi_i \uparrow \\ G_i \times_{S_i} G_i & \xrightarrow[\sim]{\Psi_i} & G_i \times_{S_i} G_i, \end{array} \quad (6.7.3.1)$$

where Φ_i is obtained from Φ by the base change $S_i \rightarrow S$, and Ψ_i is the isomorphism defined on points by

$$(g, g') \mapsto (gg', g).$$

The composite $(\pi_i \times \pi_i) \circ \Psi_i$ is covering; hence Φ_i is covering by 4.2.3 (C 3). Thus Φ is locally covering and is therefore covering, by 4.2.3 (C 5). This proves (ii) $\implies$ (i).

Conversely, assume (i), and first suppose in addition that the structural morphism $X \rightarrow S$ has a section σ . By EGA I, 5.3.13, σ is an immersion. Define $H = G \times_X S$ by the following diagram, whose two squares are cartesian:

$$\begin{array}{ccccc} H & \longrightarrow & G & \xrightarrow{\text{id}_G \boxtimes \sigma} & G \times_S X \\ \downarrow & & \pi \downarrow & & \Phi \downarrow \\ S & \xrightarrow{\sigma} & X & \xrightarrow{\text{id}_X \boxtimes \sigma} & X \times_S X. \end{array} \quad (6.7.3.2)$$

Here π , $\text{id}_G \boxtimes \sigma$, and $\text{id}_X \boxtimes \sigma$ are the morphisms defined on points, for $T \rightarrow S$, $g \in G(T)$, and $x \in X(T)$, by

$$\pi(g) = g \cdot \sigma_T, \quad (\text{id}_G \boxtimes \sigma)(g) = (g, \sigma_T), \quad (\text{id}_X \boxtimes \sigma)(x) = (x, \sigma_T).$$

Then π is covering, and H is a subgroup scheme of G representing the stabilizer $\text{Stab}_G(\sigma)$ of σ (cf. I, 2.3.3). Thus, for every $T \rightarrow S$,

$$H(T) = \{g \in G(T) \mid g \cdot \sigma_T = \sigma_T\}.$$

Let G/H denote the presheaf

$$T \mapsto G(T)/H(T),$$

and let $a(G/H)$ be its associated sheaf for $\mathcal{T}$. The preceding construction gives a commutative diagram of presheaves equipped with a G -action:

$$\begin{array}{ccc} G & \xrightarrow{\pi} & X \\ \downarrow & \nearrow_{\bar{\pi}} & \\ G/H & & \end{array} \quad (6.7.3.3)$$

where $\bar{\pi}$ is a monomorphism. Since π is covering, $\bar{\pi}$ is covering as well. By 4.3.12, $\bar{\pi}$ therefore induces an isomorphism

$$a(G/H) \xrightarrow{\sim} X.$$

We have proved: if X is a G -homogeneous space such that $X \rightarrow S$ has a section σ , then X represents the sheaf quotient G/H , where $H = G \times_X S$ is the stabilizer of σ .

In general, by hypothesis there is a covering family $\{S_i \rightarrow S\}$ such that every morphism

$$X_i = X \times_S S_i \longrightarrow S_i$$

has a section σ_i . Put $G_i = G \times_S S_i$. The morphism

$$\Phi_i : G_i \times_{S_i} X_i \longrightarrow X_i \times_{S_i} X_i$$

obtained from Φ by base change along $S_i \rightarrow S$ is still covering.

The preceding argument then gives

$$X_i \simeq G_i/H_i,$$

where H_i is the stabilizer of σ_i in G_i . This completes the proof of (i) $\implies$ (ii).

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⁶⁸Editor’s note: this reference has been added.